

A Review of Control Techniques for Inverter-Based Distributed Energy Resources Applications

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Abstract: The escalating adoption of low-carbon energy technologies underscores the imperative to transition from conventional fossil fuel-dependent sources to sustainable alternatives. The expansion of Distributed Energy Resources (DERs) signifies an essential shift towards a more resilient and environmentally friendly energy landscape. However, integrating inverter-based DERs introduces challenges, particularly in system inertia and grid instability. This review delves into the critical area of inverter-based grid control strategies, focusing on the primary and secondary control mechanisms. Primary controls are investigated, including traditional droop control and low-voltage ride-through (LVRT) capability. The secondary control strategies, involving virtual impedance (VI) and load frequency control (LFC), are vital in maintaining grid stability and reliability are reviewed. The aim is to offer a comprehensive understanding of the principles, advancements, and challenges associated with inverter-based grid controls, contributing valuable insights for the seamless integration of DERs into modern power grids.

Keywords: distributed energy resources; droop control; battery energy storage; low-voltage ride-through capability; virtual impedance; load frequency control



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1. Introduction

The implementation of low-carbon energy technologies is trending upward. Conventional energy sources primarily reliant on fossil fuels have a detrimental impact on the environment and are finite, necessitating the adoption of energy transformation methods [1,2]. As a result, DERs are being utilized to an increasingly greater extent to replace conventional energy sources in the near future [3,4]. However, controlling inverter-based DERs presents significant challenges due to their lack of inertia [5]. This lack of inertia complicates maintaining stability within the power system, leading to potential instabilities and blackouts [6]. DERs are small-scale power-generation and -storage systems and can operate in various topologies such as islanded mode, grid-connected mode, and microgrids [7].

Each type of operation mode has its own control strategy. An appropriate control strategy is foundational to ensure system reliability [8]. In this context, the focus on inverter-based grid control strategies has become paramount, as they have a vital role in ensuring the seamless integration and efficient operation of DERs [9,10].

This review paper delves into the details of such control strategies, shedding light on their primary and secondary aspects, with a specific emphasis on two critical domains: primary control strategies, including traditional droop control [11] and low-voltage ride-through (LVRT) capability, and secondary control, which includes the virtual impedance and load frequency control.

The primary control strategies are foundational to the stable functioning of DERs within the grid. Classic droop control mechanisms have long been the primary control strategy, allowing for proportional power sharing among interconnected resources [12].

However, the evolution of battery technologies has advanced the way for improved droop control, assisting in a new era of enhanced accuracy and adaptability. This review explores traditional droop control and the novel advancements that leverage batteries to optimize performance. In addition, low-voltage ride-through (LVRT) capability can operate mainly in grid-connected mode and maintain the stability of the system during abnormal conditions.

The secondary control strategies are crucial in maintaining grid stability and reliability [12,13]. By restoring the frequency and voltage, the secondary control aids the microgrid in achieving worldwide controllability. The effect of the feeder impedance variation on power sharing precision makes droop control an ineffective method. Variations in feeder impedance impact system stability. A virtual impedance (VI) technique was then used to make up for this mismatch to resolve the problem [14,15]. Additionally, load frequency control emerges as a critical aspect in the broader spectrum of secondary control, addressing the intricacies of power system frequency regulation [16,17].

In each section, several papers are investigated in detail in the primary (droop control and LVRT) and secondary control (virtual impedance and load frequency control) sections. In Appendix A, the publications from 2016 for each group are indicated in Table A1. As can be seen, attention to secondary control has increased to improve the stability of the power system. For more information, Figure 1 shows the number of papers and compares different methods from 2010.

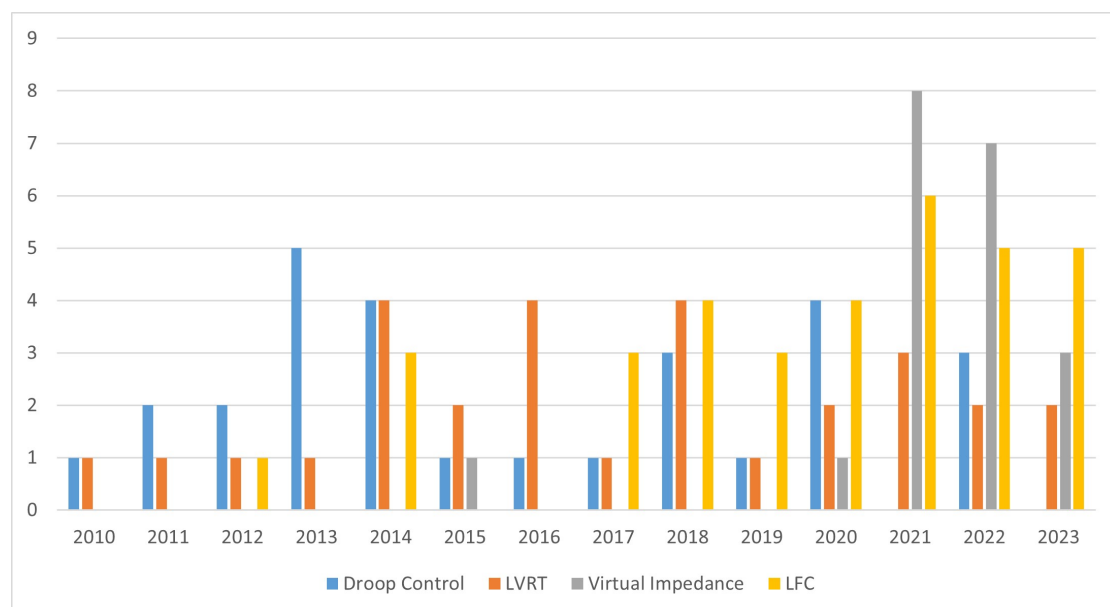


Figure 1. Comparison between different methods of primary and secondary control strategies in recent years.

Several review papers have explored various facets of this field. Ref. [18] delves into control methods specific to islanded microgrids, shedding light on vital aspects. However, a noticeable gap emerges as it primarily focuses on islanded configurations, leaving a void in the discussion of control strategies for grid-connected modes. Similarly, Ref. [19] investigates control strategies for inverter-based microgrids, but there exists a need for a more detailed exploration of each technique to provide a comprehensive understanding. The review in Ref. [20] admirably addresses various aspects of DERs' operation, comparing control strategies and studying power quality issues. However, a more comprehensive investigation into specific control methodologies could enhance the existing literature. Frequency and voltage control methods in DERs in microgrid applications are investigated in Ref. [21]. It focuses on decentralized and communication-based techniques, and a broader consideration of various methods could enrich the investigation. The critical control challenges in microgrids are illustrated in [22], and different control methods,

including droop control, virtual synchronous generators, and master–slave control, are indicated. However, various control aspects still need to be addressed.

Addressing these gaps in the existing literature, this review endeavors to navigate the detailed landscape of inverter-based grid control strategies comprehensively. In contrast to previous reviews, our primary focus is thoroughly understanding the principles, advancements, and challenges associated with primary and secondary controls. We aim to contribute valuable insights that propel the efficient integration of DERs into modern power grids, filling crucial gaps identified in prior works.

The structure of this article is as follows. Section 2 highlights the different topologies of DERs. Section 3 reviews different types of control strategies and their operating times. Section 4 presents primary control, including various types of droop control as a conventional method and improved control strategies using battery energy storage. It also covers the low-voltage ride-through capability (LVRT), grid code requirement, and LVRT for wind farms and PVs. Section 5 provides an overview of secondary control consisting of virtual impedance and load frequency control. Various methods have been investigated and compared. Key findings and potential future research are summarized in Section 6. Finally, this paper is concluded in Section 7.

2. Different Topology of DERs

DERs are small-scale power-generation and -storage systems distributed throughout a local area, such as homes, businesses, or industrial facilities. These resources can include renewable energy sources like solar panels and wind turbines and energy-storage systems like batteries and backup generators. They have various topologies and can operate in islanded, grid-connected, and microgrid modes.

2.1. Grid-Connected Mode

In regular operation, DERs are typically connected to the larger electrical grid, and they both consume electricity from the grid and export excess electricity back to it. This connection allows for efficient energy sharing and distribution. The host grid regulates frequency and voltage in grid-connected mode. In this mode, DERs usually operate in a frequency control mode that maintains a constant frequency while permitting slight variations in voltage due to variations in loads. Grid-connected mode crates a collaborative and adaptive relationship between DERs and the larger electrical grid. This integration optimizes the utilization of DERs and contributes to the stability and efficiency of the entire power system, contributing to a more sustainable and reliable energy landscape [23]. Figure 2 provides a schematic of grid-connected mode, in which the DERs are connected to the grid through the converter/inverter, transmission lines, the circuit breaker, and transformers.

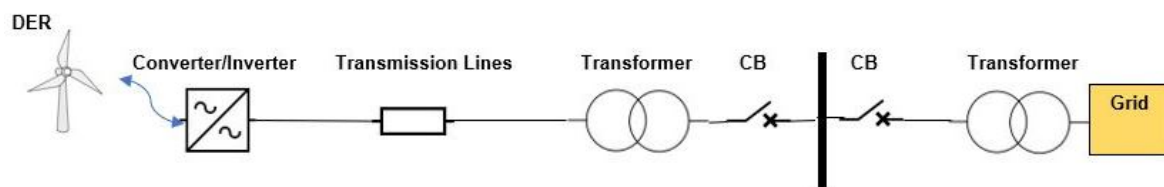


Figure 2. Grid-connected mode.

2.2. Islanded Mode

When there is a disturbance or outage on the main grid, the DERs can be configured to operate in islanded mode. In this mode, they disconnect from the main grid and continue to supply power to the local area they serve. This is achieved through specialized control systems and protection mechanisms. Within this islanded configuration, DERs commonly employ droop control mechanisms, a method that prioritizes maintaining a constant voltage while allowing for slight variations in frequency in response to changes in load. This droop control strategy ensures a stable and consistent power supply to the local area, enhancing

grid stability [23]. Figure 3 shows a schematic of islanded mode. In this mode, DERs are connected to the load through the converter/inverter, transmission lines, the circuit breaker, and transformers.

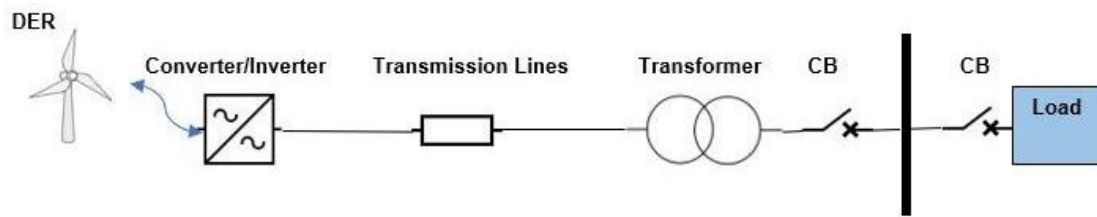


Figure 3. Islanded mode.

2.3. Microgrid Mode

A microgrid is a small-scale, local energy system that can work independently or connected to the larger electrical grid. It consists of a combination of DERs and control systems to manage and optimize energy generation and distribution. Microgrids are designed to serve a specific area or community and offer several modes of operation, including islanded and grid-connected modes. Microgrids are designed to switch between these modes seamlessly based on their specific operational requirements. They can be customized to meet the needs of the local community or facility they serve. The ability to transition between grid-connected and islanded modes balances grid support and energy independence, making microgrids a valuable tool for enhancing energy resilience and integrating renewable energy sources. Microgrid elements can interconnect via either direct current (DC) or alternating current (AC) [24]. Thanks to advancements in power electronics technology, DC energy systems have gained growing attention in recent years. DC-based energy systems offer seamless integration with DC renewable generation, energy storage, and DC electronic devices. In contrast to conventional AC systems, DC systems come with benefits such as simpler control, enhanced reliability, and greater efficiency [25]. A schematic of this mode is shown in Figure 4. In microgrid mode, AC/DC DERs and diesel generators are connected to the load and grid through the converter/inverter, transmission lines, the circuit breaker, and transformers.

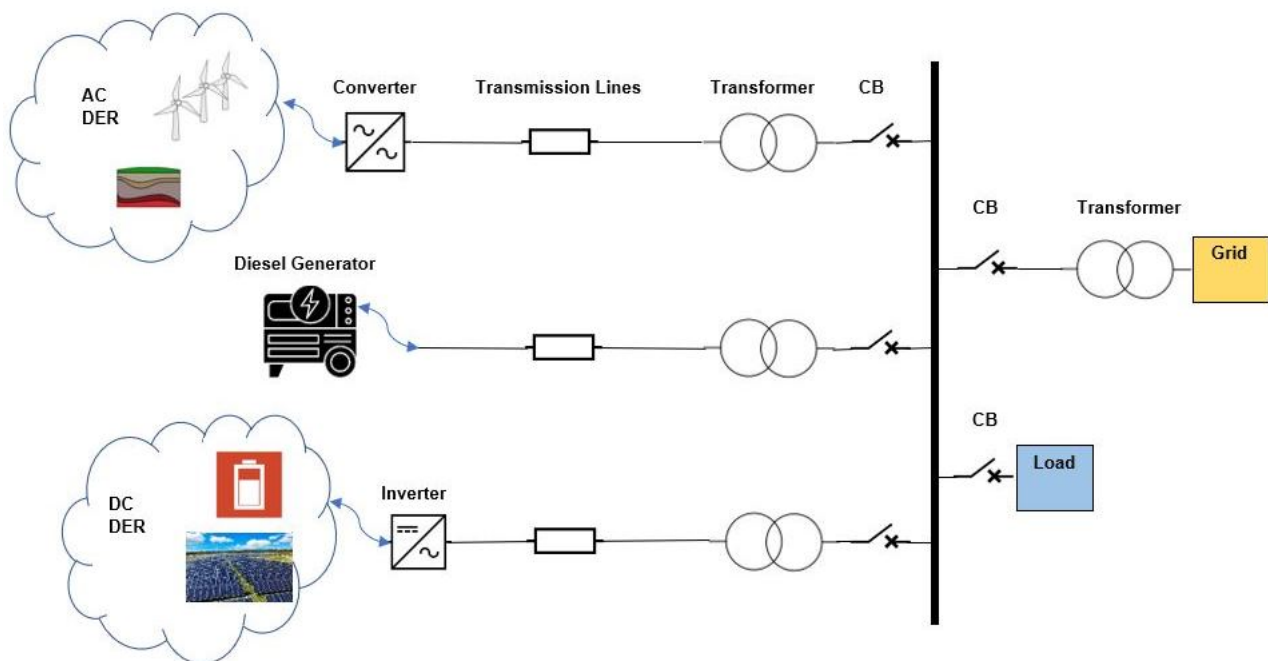


Figure 4. Microgrid topology.

3. Different Types of Control Strategies

Three distinct control strategies are employed in the power system: primary, secondary, and tertiary. Primary control operates on a fast timescale, usually within a few seconds, aiming to prevent significant frequency deviations. In contrast, secondary control restores the system in a larger timescale, contributing to the system's return to a stable, steady-state condition. Tertiary control schedules generators to adjust the power output and make economic decisions to minimize production costs while meeting demand and other operational constraints.

3.1. Primary Control Strategies

Primary control strategies are typically decentralized and executed at the device or component level. They are designed to respond quickly to local changes in conditions and demand, ensuring that the microgrid operates reliably in real time. These strategies are the foundation of microgrid operation and coordinate with secondary and central control strategies to achieve the overall microgrid objectives. Droop control methods and low-voltage ride-through (LVRT) capability are discussed as a primary control methods.

3.2. Secondary Control Strategies

Secondary control helps MGs reach global controllability and guarantees the system's operation by restoring the frequency and voltage. It coordinates with the primary control level to remove the steady-state error by providing an appropriate control signal. This control strategy can be centralized, decentralized, and distributed to optimize the operation of MGs. In this study, virtual impedance (VI) and load frequency control (LFC) are investigated as secondary control strategies in inverter-based DGs.

3.3. Tertiary Control

Tertiary control controls the power flow among its various DERs and loads in order to optimize various aspects like energy cost, the electricity market, reliability, efficiency, reserve capacity, etc. This controller optimizes the microgrid operation by producing the required information for the system operator.

Figure 5 indicates the different types of control strategies with their objectives. This paper only focuses on the primary and secondary control strategies.

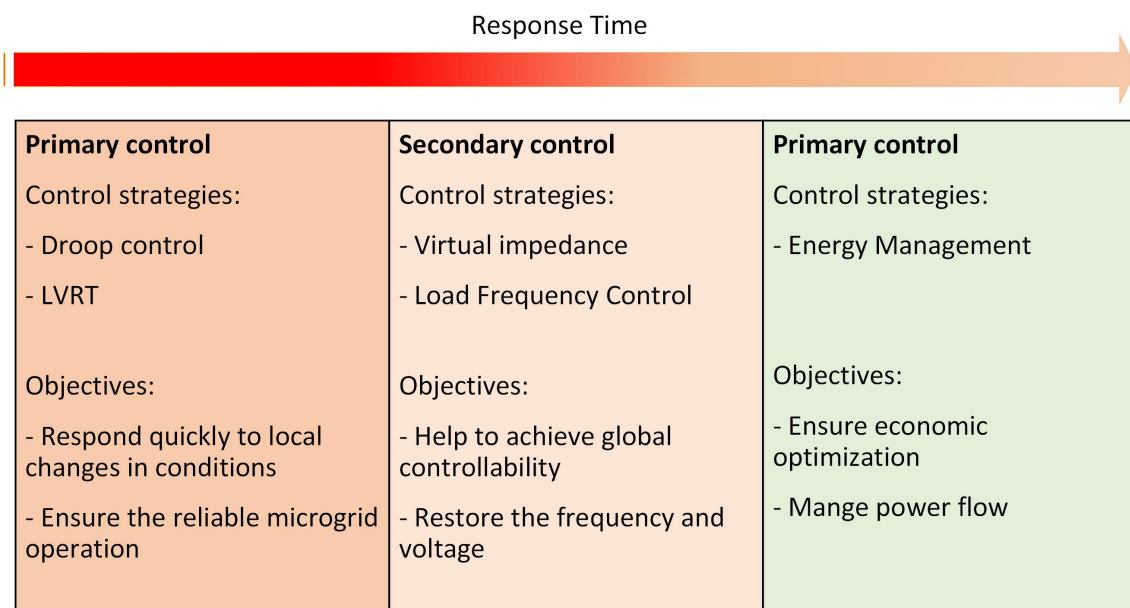


Figure 5. Different types of control strategies.

4. Primary Control

4.1. Droop Control (without Communication Methods)

A microgrid can be connected to several DERs; coordination between these units is required to fulfil the demand. Inverters, though, should meet the load requirements in stand-alone and grid-connected modes of operation and meet the voltage and frequency restrictions [26]. Various methods of control strategies have been proposed to maintain the voltage and frequency stability in the power system, in which droop control techniques are the most used strategy to regulate the exchange of active and reactive power in inverter-interfaced microgrids [27].

Droop controllers used in microgrids operate on a similar principle as synchronous generators. This method has some advantages, such as a simple implementation, adjustability by the operation parameters and for power sharing regulation, no need for any communication networks among DERs, and good reliability [28,29]. Droop control can be categorized into different control strategies: power–frequency (P/f) droop, power–voltage (P/V) droop, frequency-based signal injection, voltage-based droop (VBD), virtual flux droop, voltage–current (V/I) droop, reactive power droop control, and voltage and frequency droop control.

4.1.1. Power–Frequency (P/f) Droop

Power–frequency (P/f) droop control is an essential strategy employed in power systems to maintain the balance between power generation and consumption while ensuring stable and reliable operation [27]. It refers to the relationship between the power output and the frequency of the system. In essence, as power demand increases or decreases, the frequency of the system can deviate slightly from its nominal value. This deviation helps signal generators adjust their power output to maintain the balance between supply and demand.

$$P = P_{\text{ref}} - k_d(f - f_{\text{ref}}) \quad (1)$$

where P is the power output, P_{ref} is the setpoint for power, and k_d is the droop coefficient or gain. f and f_{ref} are the actual frequency and reference frequency.

This droop control strategy is typically used for lines that are primarily characterized by their inductance (L) or reactance (X) [30,31].

4.1.2. Power–Voltage (P/V) Droop

This droop control system relies on its resistive nature and is typically employed in resistive microgrid applications. It focuses on regulating real power flow by monitoring and adjusting the voltage levels within the microgrid. It is usually used for microgrids without rotational inertia, which are called low-voltage microgrids. P/V droop control addresses the challenges of low-voltage microgrids by allowing voltage deviations to signal the necessary adjustments in power generation or consumption. When the load increases, causing a drop in voltage, the P/V droop control system instructs the DERs to increase their power output. Conversely, if the voltage rises due to reduced load, the system signals a reduction in power generation. This real-time response helps maintain voltage stability within the microgrid, ensuring that the connected loads receive power at the desired voltage levels [32].

$$P = P_{\text{ref}} - k_v(V - V_{\text{ref}}) \quad (2)$$

where P and P_{ref} are the power output and the setpoint for power, respectively, and k_v is the droop coefficient or droop gain for voltage control. f is the actual frequency, and f_{ref} is the reference frequency [32].

4.1.3. Frequency-Based Signal Injection

Frequency deviation and power adjustment in droop control involve adjusting the output power of DERs in response to system frequency variation. When the frequency deviates, indicating a change in load or generation, droop control prompts DERs to adjust

their power output proportionally. Lower frequency leads to increased power output, while higher frequency results in decreased power output. Frequency-based signal injection is a complementary technique in droop control. It entails injecting a signal into the control system proportional to the frequency deviation. This injected signal helps DERs respond swiftly and accurately to frequency changes, improving the system's responsiveness compared to relying solely on inherent droop characteristics. The advantages of frequency-based signal injection include improved transient response, allowing the system to rapidly recover from frequency deviations, and enhanced overall system stability. The injection of a frequency-related signal enables the control system to anticipate changes effectively. The implementation of frequency-based signal injection involves control algorithms that continuously monitor system frequency, adjusting the DERs' output power accordingly.

$$P = P_{\text{ref}} - k_d(f - f_{\text{ref}}) - k_{df} \frac{df}{dt} \quad (3)$$

where P is the power output, P_{ref} is the reference, and k_d is the droop coefficient or droop gain. f and f_{ref} are the actual frequency and reference frequency, respectively. $\frac{df}{dt}$ is the rate of change in frequency, and k_{df} is the coefficient for this change or the inertia constant.

The specific algorithm may vary based on the chosen control strategy and power system characteristics [33].

4.1.4. Voltage-Based Droop (VBD)

Voltage-based droop (VBD) control is a method used in power systems to manage the output of DERs. Unlike frequency-based droop control, which responds to changes in frequency, VBD control focuses on adjusting power output in reaction to variations in system voltage. The key elements of VBD control involve adapting DER power output to maintain voltage stability within specified limits. When system voltage fluctuates, DERs modify their power output accordingly. VBD control, implemented through continuous monitoring and control algorithms, contributes to voltage stability by ensuring DERs respond appropriately to voltage variations. This method can be employed in the grid-connected and islanded mode of a microgrid. In addition, this strategy can be combined with other control methods, such as frequency-based droop control, to create a comprehensive approach to power regulation and stability in a power system.

$$Q = Q_{\text{ref}} - k_v(V - V_{\text{ref}}) \quad (4)$$

where Q is the reactive power output, Q_{ref} is the reference reactive power, and k_v is the droop coefficient or droop gain for voltage control. V is the measured voltage at the connection point, and V_{ref} is the setpoint reactive power [34].

4.1.5. Virtual Flux Droop

In essence, virtual flux droop control is a versatile strategy that considers both active and reactive power components and utilizes the phase angle and magnitude difference of the virtual flux to regulate power flow. In other words, virtual flux droop control also manages reactive power and active power control. Reactive power is essential for maintaining the voltage levels and ensuring the network operates within acceptable voltage limits. It is particularly useful in applications where precise power control is required to maintain network stability and the quality of power supply. This control method plays a vital role in modern power systems, enabling efficient and reliable operation while accommodating dynamic power generation and consumption changes. The active power and reactive power in this droop control are proportional to the flux phase angle difference and flux magnitude difference, respectively [35].

$$\delta = \delta_{\text{ref}} - K_{\delta}(P_{\text{ref}} - P) \quad (5)$$

$$|\Psi| = |\Psi_{\text{ref}}| - K_{\Psi}(Q_{\text{ref}} - Q) \quad (6)$$

δ , δ_{ref} , Ψ , and Ψ_{ref} are the actual phase angle, the nominal phase angle, the actual virtual flux, and the reference virtual flux amplitude, respectively. k_{Ψ} and k_{δ} are the droop coefficients. P_{ref} and Q_{ref} are the reference active power and reactive power, and P and Q are their actual values.

4.1.6. Voltage–Current (V/I) Droop

This approach involves adjusting the power output of DERs based on deviations in both system voltage and current from their nominal values. In V/I droop control, DERs are assigned voltage–current droop characteristics, typically represented as percentages, which determine how their power output changes in response to variations in both voltage and current. The primary objective of this control strategy is to keep the voltage and current levels within specified limits by modulating the power output of DERs. When there are deviations in system voltage and current, the V/I droop control prompts DERs to adjust their power output proportionally. This consideration of both voltage and current in the control strategy enhances load sharing among DERs, contributing to improved system stability. Implementing V/I droop control involves using control algorithms that continuously monitor both system voltage and current levels. These algorithms determine the necessary power adjustments from DERs based on observed deviations in both voltage and current. The specific algorithm and parameters may vary depending on the chosen control strategy and the characteristics of the power system.

$$V = V_{\text{ref}} - k_V(I - I_{\text{ref}}) \quad (7)$$

where V is the output voltage, V_{ref} is the reference voltage, and k_V is the droop coefficient or droop gain for voltage control. I and I_{ref} are the output current and the reference power, respectively [36–38].

4.1.7. Reactive Power Droop Control

Reactive power droop control is a technique employed in power systems to regulate the flow of reactive power, which plays a critical role in maintaining voltage levels and ensuring the stability of the electrical grid. To prevent voltage deviation, control of reactive power is required. This kind of droop control adjusts the injection or absorption of reactive power in response to changes in the system conditions. It is a vital tool to maintain the reliability and stability of the power system. It helps prevent voltage collapses, improves the performance of the grid under varying conditions, and supports the efficient operation of electrical equipment.

$$Q = Q_{\text{ref}} - k_q(V - V_{\text{ref}}) \quad (8)$$

where Q is the reactive power output, Q_{ref} is the reference reactive power, and k_q is the droop coefficient, which represents the adjustment of reactive power per unit of the voltage deviation. V is the actual voltage, and V_{ref} is the reference one [39–41].

4.1.8. Voltage and Frequency Droop Control

Voltage and frequency droop control is a comprehensive and integral method for regulating the power output of DERs within power systems. This control strategy is designed to address deviations in both system voltage and frequency, thereby contributing significantly to the overall stability and performance of the power grid. The implementation of voltage and frequency droop control relies on sophisticated control algorithms. These algorithms continuously monitor the dynamic levels of both system voltage and frequency, providing real-time data for decision-making. The control system then determines the necessary power adjustments from DERs based on the observed deviations in voltage and frequency, ensuring a timely and appropriate response to changes in the power system [23,42]. It is the combination of active and reactive power control explained in Sections 4.1.1 and 4.1.7. Table 1 summarizes the various types of droop control strategies.

Table 1. Summary of different types of droop control strategies.

Methods	Feature	Configuration (Islanded/Grid-Connected)	Ref.
Power–frequency (P/f) droop	Regulate the real power flow by adjusting the voltage level. It is used for resistive microgrid applications.	Islanded/grid-connected	[27,30,31]
Power–voltage (P/V) droop	Modify the power output by utilizing the variance in frequency. It is used for inductive lines.	Islanded	[32]
Frequency-based signal injection	Recovering the frequency through an injected signal and improving the transient response and system stability.	Islanded	[33]
Voltage-based droop (VBD)	Adjusting power output in reaction to variations in system voltage.	Islanded/grid-connected	[34]
Virtual flux droop	Control of active and reactive power.	Islanded	[35]
Voltage–current (V/I) droop	Adjusting the power output of DERs based on deviations in both system voltage and current from their nominal values.	Islanded	[36–38]
Reactive power droop control	This method controls the reactive power to prevent voltage deviation.	Islanded	[39–41,43]
Voltage and frequency droop control	Regulate the power output based on the observed deviations in voltage and frequency	Islanded	[23,42]

4.1.9. Application of Droop Control Strategy in BESSs

Batteries excel in providing primary frequency response, which is crucial for grid stability. They can instantaneously inject or absorb power in reaction to variations in grid frequency, helping to keep the frequency within the acceptable range. Traditional power plants, especially those relying on large rotating generators, might have slower response times compared to the almost instantaneous response of batteries, Figure 6. This allows them to quickly inject or absorb power to address frequency deviations. In the face of sudden changes or contingencies, such as the unexpected loss of a large generator or a sudden increase in demand, batteries can act as a rapid and reliable power source. Their ability to respond within milliseconds ensures the system remains stable even in unforeseen circumstances [44]. They can play a vital role in integrating renewable energy sources into the grid. They can smooth out the intermittency of renewable energy sources, like solar and wind, by storing excess energy when it is abundant and supplying it during periods of high demand or low renewable output. This enhances the reliability and predictability of renewable energy contributions to the grid [45]. Batteries can help manage peak demand by charging during times of low demand and discharging during times of high demand, reducing stress on the grid during peak hours [46].

Ref. [47] introduces and assesses a cooperative control strategy for an energy storage system (ESS) and micro sources in islanded operation, focusing on primary control for frequency and voltage. The ESS, through constant frequency and constant voltage control, acts as the primary controller, and a secondary control in the microgrid management system ensures that the power output of the ESS returns to zero. During grid-connected operation, micro sources and the ESS operate in power quality (PQ) control mode, with the MMS providing power output set points. In islanded operation, the upper grid operation controller adjusts microgrid frequency and voltage, addressing the challenges of power imbalance during islanding. The paper highlights the rapid response of the ESS inverter controller in milliseconds, which is crucial for maintaining frequency and voltage within normal values during islanded operation.

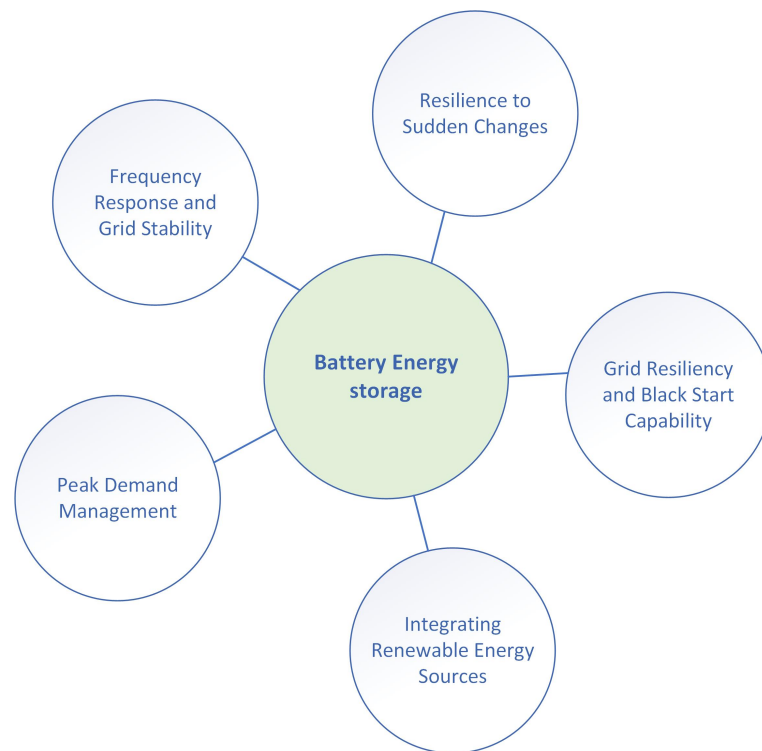


Figure 6. Contribution of battery energy storage in the primary control strategy.

The impact of integrating BESSs on short-term frequency control in autonomous microgrids (MGs) is examined in Ref. [48], particularly focusing on systems with classical and renewable energy sources (RESs). During MG islanded operation, local generators, especially RES-based ones, face limitations in grid frequency support due to a lack of controllability and energy storage capabilities. The authors propose an original BESS control structure that combines inertial response and an adaptive droop characteristic to enhance short-term frequency stability, particularly addressing battery state-of-charge limitations. This method is effective for islanded mode with limited renewable energy resources. Frequency controllers would be challenging, with a high penetration of renewable energy resources (RESs) in autonomous microgrids.

To address these challenges, Ref. [49] introduces an enhanced control method for BESSs that supports microgrid frequency, incorporates voltage support, and can disconnect to operate in islanded mode for local consumers. During primary frequency control, the proposed frequency controller combines droop control with an inertia emulation function to manage BESS active power transfer. The BESS is designed to compensate for power absorbed by local loads, improving microgrid frequency response. The BESS management involves optimization, protection actions, and interaction with higher level hierarchical control. The paper discusses various control levels, including power curtailment mechanisms to avoid battery overcharge or deep discharge. The BESS can operate in grid-connected (G-mode) or islanded mode (I-mode), with a seamless transition facilitated by a unique control method. The BESS acts as a smart load by compensating local loads, contributing to microgrid stability. In some studies, the state-of-charge (SOC) recovery is investigated in addition to frequency control. The authors in Ref. [50] focus on a battery aggregator overseeing distributed BESSs for providing primary frequency control (PFC) services in the ancillary service market. They propose a profit-maximizing coordination strategy, addressing two operational phases: frequency regulation and state-of-charge (SOC) recovery. An online optimal coordination algorithm is introduced for the frequency regulation phase, minimizing regulation failure penalties without requiring knowledge of future information. In the SOC recovery phase, the authors wanted to determine the best SOC targets for every BESS, ensuring minimal expected regulation failure penalties. The main contribution lies in

the joint optimization of BESS operation during both phases to maximize the aggregator's profit. The proposed algorithm for frequency regulation was shown to achieve optimal performance without foreknowledge. A decision tree-based SOC recovery strategy is presented, providing state-invariant and separable target SOC ranges for each BESS, allowing decentralized operation with low runtime complexity. This method cannot forecast the load and generation.

To enhance the functionality of the controller, Ref. [51] proposes a droop-type, lead-lag-controlled BESS with an adaptive state-of-charge (SOC) recovery strategy. This strategy aims to enhance system inertia, comply with regulatory requirements, and maintain flexible battery SOC based on load/PV generation forecasts and future events. This study addresses the challenge posed by the diminishing system inertia caused by increasing photovoltaic (PV) penetration in power systems, affecting primary frequency control. Unlike wind turbines, PV systems lack stored kinetic energy, limiting their ability to provide under-frequency support and potentially violating regulatory requirements. The study emphasizes the crucial role of BESSs in maintaining stability and reliability with increasing PV penetration, suggesting its applicability in large-scale interconnected or small isolated power systems. Figure 1 in Ref. [51] presents the control strategy. The variables PG , PL , Hn , and f_{ref} represent the generated power, load power demand, and system inertia constant. The droop coefficients for generation systems SG1 and SG2 are denoted as D1 and D2.

Autonomous active power control for the islanded microgrid is established in Ref. [52], which enables effective use of RESs while protecting the ESS from overcharge and over-discharge circumstances. The microgrid comprises a photovoltaic (PV) and energy storage system (ESS). The proposed method utilizes local measurement, is decentralized, and does not need external communication lines. Therefore, the reliability of the system is improved due to decreasing the risk of failure in the communication system.

In addition to methods that do not need a communication network, some studies focus on control strategies with communication networks. In Ref. [53], the authors propose a new control strategy to track voltage and frequency and decrease the transients when disturbances occur. This study includes three microgrids that are connected to the IEEE 13 Busses. Two microgrids consist of a synchronous generation unit and a battery energy storage connected to the grid through an inverter. The third microgrid has only a BESS. The study initially establishes a dynamic model for a microgrid system. Four different types of controllers are derived from this model. The suggested control scheme is centralized and relies on high-bandwidth communication. However, an opportunity exists to decrease communication requirements and related costs by designing control schemes with fewer measurement signals. This controller can solve the poor transient performance of droop controllers and also their voltage and frequency deviation. This study did not consider the communication delay.

Since communication delay can affect the performance of the controller, Ref. [54] employed a centralized strategy to address communication effects and also the challenge of adapting control algorithms in power grids, specifically focusing on the primary control level where prompt action is crucial. Traditional droop control schemes, designed for specific grid impedance, often require redesign when transitioning to different power grids, incurring higher costs and effort. The authors propose a unified secondary controller using distributed control theory for various droop schemes associated with distributed battery storage in various grid conditions to address this issue. The control design follows consensus theory, originally created for multi-agent systems, aiming to synchronize energy levels, control frequency and voltage, and proportionally distribute the active and reactive powers of battery storage systems. The approach considers communication delays and provides an upper bound to ensure system stability. The study utilizes a modified IEEE 118-bus benchmark for validation and simulation, demonstrating the effectiveness of the proposed approach in supporting different droop schemes. The unified secondary control design is expected to save design efforts and costs when applied to grid-forming distributed

BESSs in diverse grid conditions, offering potential applications in ancillary grid services such as voltage/frequency regulation, power sharing, and energy synchronization for inverter-dominated grids.

Some studies used improved controllers using various control techniques. Ref. [55] focuses on the controller design and operation of a microgrid, including a battery storage system and a direct drive wind generator. The authors designed a model predictive control (MPC) technique for the AD–DC–AC converter of the wind system to optimize wind energy capture and provide the desired reactive power. An innovative supervisory controller is presented to manage the microgrid’s operation during grid-connected and islanded scenarios, addressing the challenges of intermittent wind speed and load changes. The coordinated controller, designed using MPC, aims to mitigate active and reactive power disturbances. Notably, it emphasizes the importance of battery systems in providing dynamic reactive power support to address voltage stability issues, especially in remote wind farm locations. The paper introduces a mode where the battery shifts to voltage control mode in conjunction with the wind converter to offer dynamic reactive power support when needed. The proposed control technique aims to enhance microgrid performance by ensuring sustainable power supply and dynamic reactive power support and reducing active power oscillations. In addition, a new controller using Finite Control Set-model predictive control (FCS-MPC) for primary frequency response has been implemented in the presence of battery energy storage, as detailed in Ref. [56]. The microgrid comprises a Doubly Fed Induction Generator (DFIG) and a BESS. The rotor-side converter effectively manages wind turbine operation to maximize power extraction. The BESS, on the other hand, generates the necessary power to regulate frequency, with the specific power value determined through FCS-MPC by adjusting the droop gain. Notably, this controller operates without requiring communication and functions at the Maximum Power Point (MPP) extraction. This approach enhances the efficiency of the microgrid system.

Operating costs are considered along with the effect of the battery on primary control. Ref. [57] introduces a two-level profit-maximizing strategy for BESS owners participating in the primary frequency control (PFC) market. The strategy involves planning and control, with the optimal BESS control minimizing operating costs by maintaining the optimal range. It proves that the optimal BESS control is a “state-invariant” strategy, allowing for offline computation with low complexity. Additionally, the minimum operating cost is a decreasing convex function of the BESS energy capacity, guiding the optimal BESS sizing that manages the equilibrium between capital investment and operating expenses. The theoretical framework provides insights into maximizing economic benefits in ancillary service markets, contributing a valuable perspective to complement simulation-based studies. The study’s main contributions include reducing the complexity of the optimal BESS control problem and demonstrating the connection between operational cost and BESS energy capacity. The paper lays a foundation for understanding optimal planning and control approaches for BESSs in PFC markets, offering theoretical insights beyond simulation-based approaches. A summary of improved control strategies using battery energy storage is indicated in Table 2.

Table 2. Application of droop control strategy in BESSs.

Ref.	Feature	Type of Controller (Centralized/Decentralized)	Islanded/ Grid-Connected
[47]	ESS operates in power quality (PQ) control mode in grid-connected mode and regulates microgrid voltage and frequency and during islanding.	Decentralized	Islanded/ grid-connected
[48]	Proposes a novel BESS control scheme that combines an adaptive droop characteristic and inertial response.	Decentralized	Islanded

Table 2. Cont.

Ref.	Feature	Type of Controller (Centralized/Decentralized)	Islanded/ Grid-Connected
[49]	The controller combines droop control with an inertia emulation function to manage BESS active power transfer during primary frequency control.	Decentralized	Islanded/ grid-connected
[50]	The controller aims to regulate frequency and recover the SOC.	Decentralized	Islanded
[51]	A droop-type, lead–lag-controlled BESS is designed with an adaptive state-of-charge (SOC) recovery method.	Decentralized	Islanded
[52]	Autonomous active power control is designed to enhance the stability and protect the ESS from overcharging and over-discharging.	Decentralized	Islanded
[53]	It can solve droop controllers’ poor transient performance and voltage and frequency deviation.	Centralized	Islanded
[54]	It is designed for multi-agent systems, aiming to synchronize energy levels and control voltage and frequency. The communication delay is considered.	Centralized	Islanded
[55]	Proposes a model predictive control (MPC) method for the AC–DC–AC converter to capture and provide the desired reactive power.	Decentralized	Islanded/ grid-connected
[56]	Employed a Finite Control Set-model predictive control (FCS-MPC) for primary frequency response by adjusting the droop gain	Decentralized	Islanded
[57]	The strategy involves planning and control, with the optimal BESS control minimizing operating costs by maintaining the SOC within an optimal range.	Decentralized	Islanded

4.2. Low-Voltage Ride-Through Capability

Low-voltage ride-through (LVRT) capability is typically considered a primary control strategy, especially in grid-connected renewable energy systems, such as wind turbines and solar inverters. Primary control strategies, including LVRT, are responsible for rapid responses to disturbances and variations in grid parameters. LVRT is designed to enable grid-connected generators to remain connected to the grid and continue operating during voltage sags or disturbances, even when the grid voltage falls below normal operating levels. This capability helps maintain grid stability and reliability during adverse grid conditions. It is a crucial feature for renewable energy systems connected to the electrical grid [58].

4.2.1. Grid Code Requirement

Grid codes are technical specifications and regulations defining the requirements for connecting power generators to the electrical grid. LVRT capability is one aspect of grid codes that specifies how power generators should respond and remain connected during voltage sags or short-term disturbances in the grid [59,60]. The specific LVRT requirements can vary between different regions and countries, as they are often defined by the relevant grid operators or regulatory authorities. However, some common elements found in LVRT grid code requirements include the following:

(a) Ride-through duration: Grid codes typically specify the duration for which a power generator must ride through a low-voltage condition without disconnecting. This duration is often expressed in seconds.

(b) Voltage magnitude and variation: The grid code may define the acceptable range of voltage magnitudes and the voltage change rate (dV/dt) during a disturbance. Power generators must remain connected and continue to operate within these limits.

(c) Reactive power control: LVRT requirements may involve controlling reactive power to help stabilize the grid during and after a disturbance. Power generators are often expected to contribute to voltage support and recovery [61].

(d) Resynchronization: After a disturbance, the generator may need to resynchronize with the grid and safely reconnect without causing further issues [62].

(e) Communication and monitoring: Grid codes may include requirements for communication between the power generator and the grid operator. Monitoring capabilities are often required to provide data on the generator's performance during disturbances. Power system developers and operators must be aware of and comply with the LVRT requirements specified in the relevant grid codes to ensure the reliable and stable operation of the electrical grid. Local regulations and standards should be consulted for the specific LVRT criteria applicable to a particular location.

Figure 7 illustrates the typical specifications for LVRT capability in Germany for wind turbines (WTs). As indicated in Figure 7, if the fault duration and system voltage stay above the red line, the WTs must maintain their connection to the grid. Otherwise, they are allowed to be disconnected. This curve is different for each country, and as mentioned, it is related to the grid code definition.

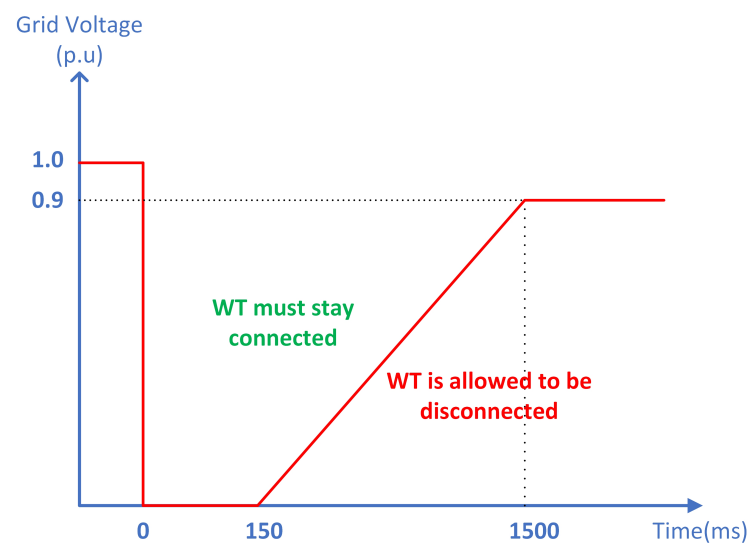


Figure 7. Ride-through curve for WTs in Germany [63].

4.2.2. LVRT Methods for Grid-Connected Wind Farms

Different methods to enhance LVRT capability can be classified into two categories: hardware modification and control system modification. Each method has a different configuration based on its location and performance. The advantages and disadvantages of these methods are compared in Table 3. These methods have different performances and can restrain the fault. The classification of each control strategy is indicated in Figure 8.

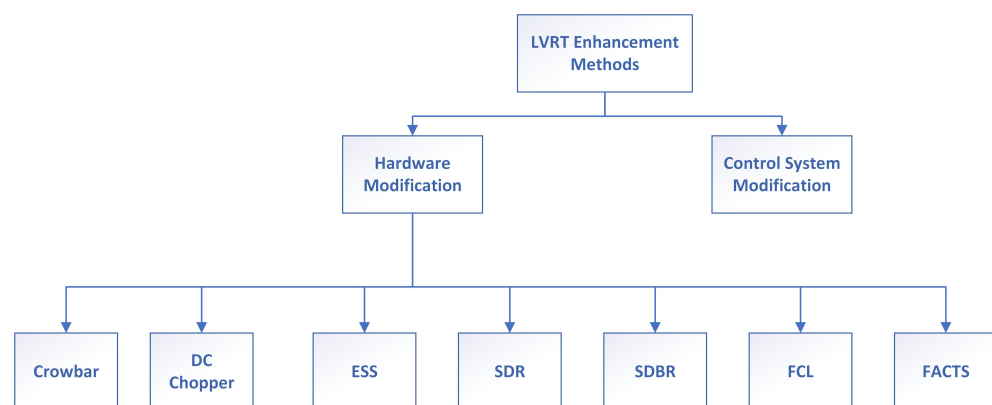


Figure 8. LVRT enhancement classification.

Table 3. Caparison between two commonly used methods to improve LVRT capability.

Methods	Advantages	Drawbacks
Hardware modification	(1) Increased robustness: To improve the system's resistance to voltage dips or disruptions, hardware improvements usually entail adding physical components or devices. As a result, the system may be more resilient to voltage fluctuations and less dependent on control mechanisms. (2) Independence: Hardware modifications can offer some protection against voltage variations, even in the event of control system problems or failures. This means that it is not dependent on a control system.	(1) Cost: Hardware modifications can be expensive, including implementing additional equipment. (2) Complexity: Adding hardware modifications could make the system more complex and possibly cause maintenance problems.
Control system modification	(1) Cost effectiveness: These methods usually entail modifying the algorithms or control techniques already employed in the system. This might be a more cost-effective option, particularly for systems already operating. (2) Adaptive and flexible: These are more adaptive and flexible. Updates and reprogramming of software provide more straightforward modifications and fine-tuning without physical changes.	(1) Dependence on control systems: They rely heavily on properly operating control systems. If there are control failures, the control system might be unable to ride through low-voltage events. (2) Limited protection: Compared to hardware modifications, control alterations may not provide as much immediate and direct protection. They may reduce issues within the system, but they might not be as useful in some extreme or abrupt voltage dip situations.

(a) Crowbar: The crowbar, a common protective system in DFIG wind turbines for LVRT, involves the installation of a series of resistors in parallel between the rotor windings and the AC side of the rotor-side converter (RSC). This mechanism activates when there is an overcurrent in the rotor or an overvoltage in the DC link. It establishes a path of low resistance, enabling a high flow of current. By closing the crowbar switch, the RSC disconnects from the rotor windings, preventing damage due to overcurrent. The precise selection of the connection of the crowbar duration and resistance holds significant importance in enhancing transient stability [64,65].

(b) DC chopper: A DC chopper consists of a resistor, insulated gate bipolar transistor (IGBT), and diode arranged in parallel with a DC link capacitor. This setup is positioned between the rotor-side converter (RSC) and the grid-side converter (GSC). The IGBT regulates the on and off cycles of the DC chopper circuit. To sustain the DC link voltage, a braking resistor is introduced in series with the IGBT. The abrupt surge in rotor current occurring during a grid fault results in an overvoltage of the DC link due to the charging of the DC link capacitor. In such instances, the IGBT needs to be active while braking resistors are employed to regulate the DC link voltage. Although a traditional DC chopper can manage the overvoltage concern, it falls short in maintaining transient overcurrent within acceptable limits on both the stator and rotor sides [66].

(c) Energy storage system (ESS): Incorporating an energy storage system into a grid-connected Wind Energy Conversion System (WECS) can even effectively smooth power output and notably enhance power quality. Moreover, a WECS outfitted with an energy storage system can operate independently and contribute to maintaining grid frequency. Despite the enhancements to power quality and system stability achieved through the utilization of an energy storage system, the costs associated with manufacturing substantially escalate [67].

(d) Series dynamic resistor (SDR): This approach involves using a dynamic resistor, which is placed in series between the rotor windings and the rotor-side converter (RSC). Power electronic switches are responsible for managing this SDR. During normal operating conditions, the switch remains closed, bypassing the resistor. However, when a fault occurs, the switch opens, activating the series resistor between the RSC and rotor windings, allowing the rotor current to flow through it. The utilization of the SDR serves multiple purposes. It effectively regulates rotor overcurrent, subsequently reducing the charging current directed to the DC link capacitor. By diminishing this current, the likelihood of

DC link overvoltage is minimized. Moreover, integrating the SDR results in an extended operational duration of the RSC, enabling the injection of more reactive power into the grid to facilitate system recovery. This extended operation can improve the system's capability to withstand faults without frequently resorting to the use of a crowbar short-circuit, thus enhancing the ability of the system to ride through faults. Consequently, the presence of the SDR enables the RSC to remain operational during fault conditions [68].

(e) Series Dynamic Braking Resistor (SDBR): The primary objective of the SDBR is to stabilize active power in the event of faults. It consists of a resistor that runs parallel to a switch. When a fault arises, introducing a resistor into the generation circuit causes a rise in the generator's terminal voltage. Consequently, this action helps alleviate the destabilizing reduction in electrical torque and power. The placement of the SDBR varies depending on factors such as available space and switching costs, among others. For example, the SDBR can be centrally located or distributed among individual wind turbines [69].

(f) Fault current limiter (FCL): The FCL can limit the fault current and prevent the disruption of the protective device. In fact, they are utilized in power systems to protect wind turbines and also transmission lines and generators. Since the FCL can inject the resistance into a stator circuit, it can regulate voltage during fault conditions. The FCL has different configurations [70].

(g) Flexible Alternating Current Transmission Systems (FACTSs): This approach falls into two primary categories: the Static Var Compensator (SVC) and the Static Synchronous Compensator (STATCOM). The SVC involves a combination of a thyristor-controlled reactor and a thyristor or mechanically switched capacitors. Its role in enhancing the transient stability of DFIG wind turbines lies in injecting reactive power into the grid. Additionally, it serves to dampen power oscillations in transmission lines.

On the other hand, the STATCOM, a reactive power device, operates on power electronic principles utilizing a voltage source converter. The STATCOM's control is managed through digital methods, and it is connected in parallel (shunt) to the power system through an R-L filter and a transformer. Its configuration is contingent upon the voltage level and typically employs a two- or three-level voltage source converter.

A key advantage of the STATCOM over the SVC is its ability to compensate for current independent of the voltage level at the Point of Common Coupling (PCC). Both methods, STATCOM and SVC, are effective in various conditions, whether during regular system operation or in the event of faults. During normal conditions, these systems contribute to voltage regulation by injecting or absorbing reactive power, thereby enhancing overall system stability [71,72].

The Unified Power Flow Controller (UPFC) is a sophisticated FACTS device that integrates the capabilities of both the SVC and STATCOM. It enables simultaneous management of both active and reactive power flows within the transmission line. The UPFC plays a significant role in enhancing system stability and controlling voltage levels in LVRT events. Hardware modification methods and their location in DFIG wind farms are indicated in Figure 9. These hardware devices are shown as dotted lines. This means that they are added to the system based on their application and control strategy. Table 4 summarizes the various types of hardware modification methods for wind farms.

(h) Control system modification methods: Control system modification methods for LVRT in wind turbines involve adjusting the control strategies and parameters to improve the performance of the turbines during grid voltage disturbances and faulty conditions. These methods aim to improve the capability of wind turbines to remain connected to the grid and recover from low-voltage events, thereby ensuring grid stability and reliability. There are various types of modifications, and the specific modifications depend on the turbine design, control system architecture, and characteristics of the grid to which the turbines are connected.

Ref. [73] designs a fuzzy logic controller to improve the LVRT for PMSG wind turbines and energy storage systems. This study addresses the challenges of LVRT in WTs by proposing an enhanced coordinated LVRT control method. The traditional approach

focuses on DC link voltage regulation using energy storage systems (ESSs). Still, this work introduces a fuzzy logic algorithm considering rotor inertia, pitch angle control, and ESS SOC. By incorporating pitch angle control at high wind speeds, the proposed method provides a more stable LVRT response and increases reserve power during grid faults.

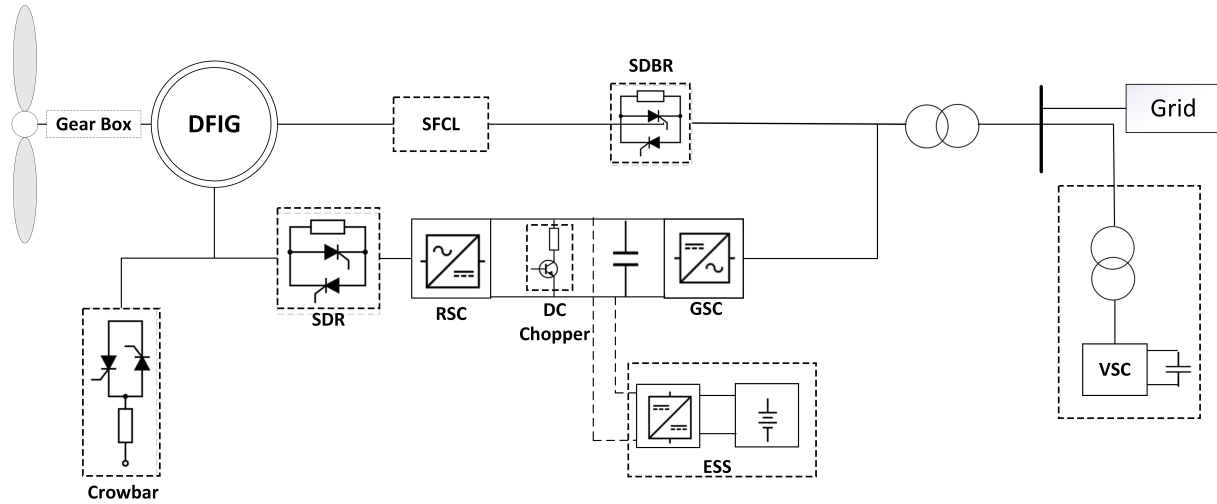


Figure 9. Hardware modification for LVRT enhancement and their location in DFIG wind farms.

Table 4. Summary of hardware modification methods for wind farms.

Methods	Feature	Ref.
Crow bar	A series of resistors in parallel between the rotor windings and the AC side of the rotor side; it is activated during overcurrent in the rotor or overvoltage in the DC link.	[64,74]
DC chopper	It consists of a resistor, IGBT, and diode arranged in parallel with a DC link capacitor. It is positioned between the RSC and GSC. During a grid fault, the IGBT is activated while braking resistors are employed to regulate the DC link voltage.	[66]
Energy storage system	It can smooth power output and contribute to maintaining grid frequency.	[67]
Series dynamic resistor (SDR)	It uses a dynamic resistor and a switch, which is placed in series between the rotor windings and the rotor-side converter (RSC). The switch is closed during normal operation and is open during a fault to activate the series resistor between the RSC and rotor windings to regulate rotor overcurrent.	[68]
Series Dynamic Braking Resistor (SDBR)	It stabilizes active power during fault conditions. It consists of a resistor that runs parallel with a switch. It is inserted when a fault occurs to prevent destabilization of the system.	[69]
Fault current limiter	The FCL can inject resistance into a stator circuit and regulate voltage during a fault.	[70]
FACT devices	It includes two methods: the Static Var Compensator (SVC) and the Static Synchronous Compensator (STATCOM). These methods contribute to voltage regulation by injecting or absorbing reactive power, enhancing overall system stability.	[71,72]

A new communication-independent LVRT control technique is introduced in Ref. [75] for high-voltage direct current (HVDC)-connected offshore wind farms (OWFs). The proposed strategy ensures compliance with LVRT requirements and active power recovery ramp rate limits and addresses the Voltage-Dip-Induced Frequency (VDIF) issue during fault conditions. By enhancing active power injection during LVRT, utilizing HVDC link capacitors, and coordinating controls at the Wind Turbine Generators (WTGs), grid-side converters (GSCs), Static Excitation Controllers (SECs), and Reactive Power Controllers (RECs), the proposed strategy outperforms conventional and modified conventional approaches. There are some machine learning techniques to improve LVRT. A reinforcement learning technique is employed in Ref. [65], specifically Q-learning and dynamic fuzzy Q-learning (DFQL), for controller design to enhance the LVRT capability of a hybrid power

system utilizing a Convertible Static Compensator (CSC). Different CSC configurations are considered, including a Static Synchronous Series Compensator (SSSC), one STATCOM, two STATCOMs, and a Unified Power Flow Controller (UPFC). Simulation results reveal that the DFQL-based controller outperforms classical PID and Q-learning controllers in improving LVRT capability. Additionally, the UPFC and two STATCOM configurations of the CSC demonstrate superior voltage support. The research explores the potential of reinforcement learning controllers in conjunction with the CSC for achieving LVRT capability in a hybrid power system with synchronous and asynchronous generators.

4.2.3. LVRT Methods for Grid-Connected PVs

Various LVRT methods have been proposed for PV applications. Most of the studies focus on reactive power injection and advanced control strategies. This can be categorized into the following methods:

(a) Reactive power injection: In PV systems, this is a vital mechanism for grid support, voltage stability, and compliance with grid codes, significantly enhancing LVRT capabilities. Ref. [76] explores reactive power injection (RPI) strategies for single-phase PV systems, presenting possibilities such as constant average active power and thermally optimized control. These strategies comply with existing grid codes and are demonstrated through simulations and experiments on a 1 kW system in LVRT mode, showing effectiveness in grid support. The compliance of proposed strategies with medium-voltage grid codes and their potential to improve reliability during LVRT operation are emphasized, offering design guidelines for PV inverter development to meet future grid demands, especially in low-voltage grids during normal operation and grid faults.

(b) Energy storage system: These enhance LVRT in PV systems by providing rapid and flexible support during grid disturbances. The ESS contributes to voltage stability through a fast response, active and reactive power injection, and smooth transition between grid-connected and islanded operation modes. Additionally, the ESS aids in smoothing voltage fluctuations, complying with grid codes, and ensuring consistent power output. Overall, integrating an ESS with PV systems enhances grid reliability and stability during normal and abnormal operating conditions. A super-capacitor control strategy for optimizing energy storage is presented in [77]. It introduces a solution involving a single-phase connected PV array with a super-capacitor. It focuses on investigating single-phase inverters, specifically considering the impact of grid faults on the DC side dynamics and the Maximum Power Point Tracking (MPPT) algorithm.

(c) FACT devices: Typically, these have fast response times on the order of milliseconds. They quickly respond to changes in the grid voltage, providing immediate support during voltage sags. They offer dynamic control over reactive power, allowing them to regulate voltage and stabilize the system during variations caused by events like sudden changes in load or faulty conditions. These devices can be integrated with smart inverters, combining the advantages of both technologies. Smart inverters can focus on real power generation, while FACTS devices handle reactive power support and voltage regulation. This technique is employed in Ref. [78], which introduces a novel concept called PV-STATCOM, a smart inverter control system for photovoltaic (PV) solar farms. Unlike traditional smart inverters, the PV-STATCOM operates as a dynamic reactive power compensator (STATCOM) on a 24/7 basis, providing continuous voltage control. At nighttime, the entire inverter capacity is dedicated to STATCOM operation, and during daytime disturbances, the smart inverter temporarily suspends real power generation to release its full capacity for STATCOM operation, ensuring critical grid support. The response time of the PV-STATCOM is comparable to an actual STATCOM (1–2 cycles). The proposed PV-STATCOM remains stable during transitions between different modes, which is achieved through the appropriate design of PI controllers.

(d) Control strategies: The main aim of control strategies in PV applications for LVRT is to enhance the capability of PV systems to maintain stable operation during periods of low grid voltage. These strategies focus on implementing effective techniques to inject reactive power, manage power electronics, and control the system's behavior to ensure resilience

against voltage disturbances, such as grid faults or drops. The goal is to improve the system's ability to ride through and quickly recover from low-voltage events, contributing to grid reliability and stability.

A multi-mode operation strategy is proposed in Ref. [79] for a three-phase PV power system with LVRT capability. The proposed control strategy involves an interleaved boost converter that enables continuous active power extraction from PV arrays during LVRT. The system operates in three modes: Maximum Power Point Tracking (MPPT), Constant Power Control (CPC), and Short-Circuit Current (SCC). The interleaved boost converter ensures MPPT operation during slight voltage sags, triggering CPC mode when PV array power exceeds the allowable inverter power. The system transitions to SCC mode in a severe grid fault, preventing circuit damage with a derived incremental duty cycle. Additionally, a Current Amplitude Limitation Control (CALC) approach for the three-phase inverter optimizes reactive power output within the rated current amplitude during voltage dip, meeting various LVRT codes.

Ref. [80] introduces a novel application of a Continuous Mixed ℓ_2 -Norm (CMPN) algorithm-based adaptive control strategy to enhance the LVRT capability of grid-connected photovoltaic (PV) power plants. The control strategy includes a Maximum Power Point Tracking (MPPT) operation using the boost converter and a vector control scheme for the grid-side inverter. The CMPN algorithm-based adaptive proportional–integral (PI) controller is employed for its fast convergence, updating controller gains online without the need for fine-tuning. This control method is evaluated with a Taguchi-approach-based optimal PI controller in scenarios involving symmetrical and unsymmetrical faults along with instances of circuit breakers failing to close again due to a permanent fault. The results show that the CMPN algorithm-based adaptive control strategy yields faster, better-damped, and superior system responses compared to the Taguchi-approach-based optimal PI control scheme under various fault conditions.

A model predictive control scheme with a low switching frequency for effective control is developed in Ref. [81] to effectively control active and reactive power exchange between the PV system and the grid. This control strategy ensures balanced grid currents during disturbances, meets LVRT regulations, and achieves optimal active and reactive power regulation under transient conditions. A three-phase Module-Integrated Converter (MIC) is employed, and the discrete-time model of the MIC is developed in a synchronous reference frame to predict future current control values and switching states. The proposed predictive switching algorithm minimizes a cost function to control injected active and reactive power. An islanding detection method is designed in Ref. [82] for a two-stage PV inverter that operates independently of the instantaneous grid voltage. Instead, it detects islanding conditions through the saturation of the proportional–integral (PI) controller in the outer voltage control loop. The proposed method is integrated with a simple and accurate implementation strategy for LVRT operation. LVRT functionality is implemented to handle low-voltage conditions without unnecessary interruptions to the local load. The paper discusses the LVRT method, which allows the inverter to ride through short-duration low-voltage conditions, but initiates anti-islanding if the low-voltage condition persists. The integrated control algorithm is versatile, catering to standalone and grid-connected modes with seamless transitions, incorporating islanding detection, LVRT features, and Maximum Power Point Tracking (MPPT).

The challenge of active power back-flow in three-phase cascaded H-bridge (CHB) PV grid-tied inverters during LVRT poses a risk of system imbalance and potential shutdown. Addressing this, Ref. [83] proposes an optimized LVRT control strategy for cascaded modular medium-voltage PV power-generation systems. This strategy employs different control methods to prevent active power back-flow in three types of asymmetric grid drop conditions, including a single-phase short-circuit fault with the ground, a two-phase short-circuit fault with the ground, and a two-phase short-circuit fault without the ground, enhancing adaptability to varying output powers and grid voltage failures, consequently improving the LVRT capability of the system. Table 5 gives a summary of LVRT methods for PV.

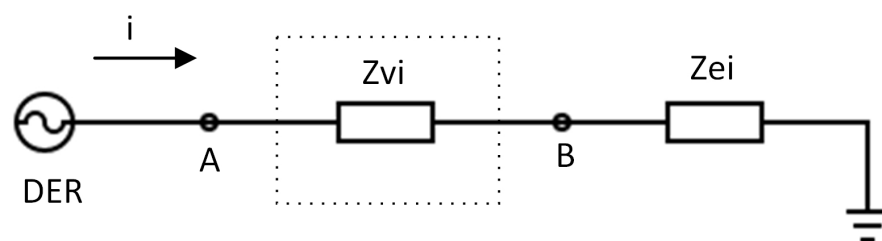
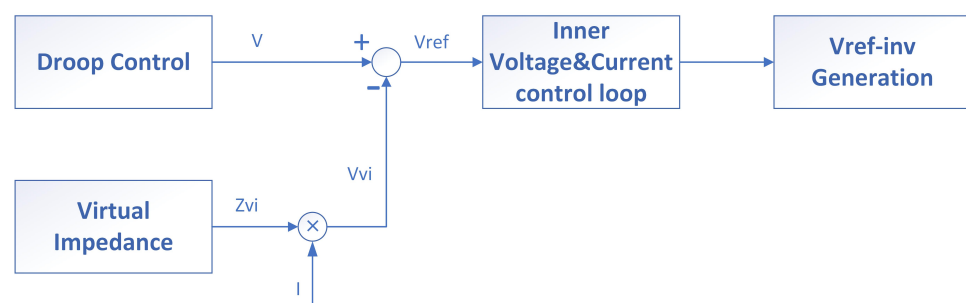
Table 5. Summary of modification methods for PVs.

Methods	Feature	Ref.
Reactive power injection	Injecting the reactive power into the grid enhances the system's ability to maintain voltage stability and support the grid during voltage sags or faults.	[76,84]
Energy storage system	This method has a fast response and can provide reactive power to keep voltage levels within acceptable thresholds during grid disturbances and inject active power during low-voltage events.	[77]
FACT devices	These devices have a rapid response and allow immediate support during voltage sags in the grid; dynamic control over reactive power enables voltage regulation and system stabilization, and they are effective in addressing variations caused by sudden load changes or faults.	[78]
Control method strategies	These strategies focus on implementing effective techniques to inject reactive power, manage power electronics, and control the system's behavior to ensure resilience against voltage disturbances, such as grid faults or drops.	[79–82,85]

5. Secondary Control

5.1. Virtual Impedance

Due to the penetration of RESs in power systems, the stability of the power system has been affected. Since conventional power systems have rotational inertia, they respond better to fast load changes to maintain stability, whereas RESs are inverter-based sources with no rotational inertia. Thus, the control strategy becomes essential to maintain system reliability [26]. Droop control techniques are the most often utilized strategy for managing the exchange of active and reactive power in inverter-interfaced microgrids. This method has some advantages, such as a simple implementation, adjusted by the operation parameters and for power sharing regulation, not needing any communication networks among DERs, and finally, good reliability. Despite its advantages, this method has some disadvantages. Since the feeder impedance varies, droop control is not an effective strategy due to the impact of this variation on power sharing accuracy [86]. Feeder impedance variation affects system stability. Hence, to solve this issue, a virtual impedance (VI) strategy was employed to compensate for this mismatch. As shown in Figure 10, VI can manage this mismatch by changing Z_{vi} . This VI can change the reference voltage for inverters, as indicated in Figure 11 [87]. It can modify the reference voltage to mitigate the effect of impedance mismatch.

**Figure 10.** Schematic of VI in the power system.**Figure 11.** Schematic of VI in terms of control strategy.

Many methods for VI have been established to modify the reference output voltage of the voltage source converter [88,89]. Various types of VI have been suggested, such as resistive, capacitive, inductive, or complex virtual impedance. An adaptive virtual-impedance-based droop control strategy is developed in Ref. [90] to address the reactive power sharing issue caused by the variation in the line impedance. The virtual impedance, which is inductive and without any resistors, can be determined by local information and does not need extra measurement devices, which would decrease the complexity of the VI (using the reactive power and the output line current as inputs for the proposed VI). This model is investigated in islanded mode with three parallel DGs. A small signal stability analysis of the controller is performed to obtain the optimal control parameters. In Ref. [91], the coordinated virtual impedance is formulated using the complex virtual impedance approach, including inductance and resistance. The inductance and resistance compensate for a reactive power error and an active power error, respectively, to achieve accurate power sharing irrespective of the microgrid system parameters. Both virtual resistance and virtual inductance are simultaneously tuned to compensate for the variation in line impedance among DGs, and the microgrid system's stability is improved by increasing damping for the whole system. The coordinated virtual impedance controller consists of the proposed controller, the droop controller, and the inner voltage and current loops. The controller design process and theoretical analysis of the whole system are presented in detail, employing the small-signal analysis.

Inverter current feedback (ICFB), Capacitor Current Feedback (CCFB), Capacitor Voltage Feedback (CVFB), Grid Current Feedback (GCFB), capacitor voltage feedforward (CVFF), and PCC Voltage Feedforward (PVFF) are the six kinds of commonly used active damping methods. In Ref. [92], a generalized virtual impedance model for these widely used active damping methods is suggested and aims to decrease the complexity of the control system and computation time delay in controllers. A hybrid active damping strategy is implemented, including inverter current feedback and capacitor voltage feedforward. Since this method utilizes the capacitor voltage and inverter current, it does not need more sensors. It decreases the cost of the control strategy and also its failure. The controller was simulated and tested in grid-connected mode. This article focused on grid-connected mode; however, the isolated mode, considered a weak grid, was not investigated.

In Ref. [93], the authors introduced a virtual dynamic grid impedance, which varies based on the quantity of inverter-based resources (IBRs) linked to the grid. The focus is on frequency spectrum analysis to investigate the harmonic impact at the PCC point using the electromagnetic transient (EMT) simulation model of grid-connected inverter-based resources. This model investigates the harmonic analysis by employing the EMT model in IBR plants based on filters such as L, LC, and LCL. Another harmonic analysis is investigated in Ref. [94], and an adaptive decentralized technique is modeled to adjust the virtual impedance within the controller of a DG according to its output current. The range for parameter stability is validated through small signal eigenvalue analyses. The suggested method does not need any extra measurement systems and is adopted with two different control methods: conventional droop (to solve the reactive power sharing issue) and inverse droop control (to solve the active power sharing issue). This study demonstrates various scenarios.

Moreover, most of the mentioned studies on reactive power sharing focus on parallel microgrids that are ineffective for multi-bus radial MGs. Ref. [88] presented an adaptive VI in hybrid AC/DC microgrids in parallel and radial configurations to tackle this issue. The control method adjusts the output impedance by monitoring the difference in active and reactive power between the inverter terminal and the microgrid connection point. In addition, the stability of low-voltage resistive microgrids and medium-voltage inductive microgrids is investigated.

Virtual flux droop (VFD) is another droop control strategy method integrated with VI in some studies. To compensate for the mismatched line impedance and power sharing, an adaptive virtual flux droop is deployed in Ref. [95], and it is suggested to use it with a virtual impedance idea simultaneously. In other words, this model is constructed based on

the direct flux control strategy. In the VFD method, the phase angle and magnitude of the virtual flux are modified instead of the frequency and voltage amplitude. A small-signal approach is employed to adjust the control parameters. However, implementing the VFD method is difficult in a large system and has slow dynamic performance.

In addition to droop control methods that do not require any communication networks and utilize local information to build a controller, some droop control methods require extra sensors. A consensus virtual output impedance control is one of the communication-based controllers illustrated in Ref. [96] for the radial microgrid. This research shows that only virtual resistance or virtual reactance is sufficient, and the model just needs tuning of the coupling gain and does not rely on the tuning of PI controllers. To show the effectiveness of the control strategy, it is tested in steady state, transients (load up-stepping), plug-and-play, and communication delay. Another consensus control method in radial microgrids is developed in Ref. [97] to tune the virtual output impedance. The proposed method uses the reactive power measurement data of each DER and the primary control information of other DERs to find the value of the mismatched impedance and then update the impedance as an input for a virtual voltage drop. Finally, the voltage reference of the inverter would be generated by using a voltage and current control as an inner control. This method concludes the reactive power correction. It is shown that only dynamic virtual output impedance is required, the static component can be nullified, and virtual reactance has a quicker correction dynamic compared to the virtual resistance. The structure of the controller is not complex, and since the PI controller is not utilized, this leads to a simpler tuning. But, as mentioned, consensus control needs communication networks. So, this method is expensive, and there are limitations to DG expansion and flexibility.

The power quality problem is addressed in Ref. [98], and it focuses on the stability margin. For this purpose, a bifurcation analysis approach is employed to analyze the effect of the parameter changes on the stability margin. To perform this analysis, the dynamic model of a microgrid consists of the controller, networks, and load implemented. Droop tuning procedures may cause a flicker phenomenon, which can be avoided by deploying VI and solving the instability problems. To solve the unbalanced power sharing issue, an adaptive technique for adjusting the negative sequence (NS) virtual impedance of the DG is designed in Ref. [99] for islanded AC microgrids. VI is implemented based on its output current and integrated with the inverse droop techniques and the droop control for the positive sequence. Small-signal analysis is utilized to find the stable range of the control parameters. To balance the inverter output currents under unbalanced AC-side conditions, a method with detailed analysis is illustrated in Ref. [100]. The strategy is founded on the idea of a virtual impedance that is placed in series with the inverter output. The effect of virtual impedance for steady state and transient conditions is investigated. Then, different types of virtual impedance combinations are implemented and compared to the results to choose the best performance. This method has a fast settling time and decreases the overshoot compared to the conventional ones. In conventional droop control, the droop coefficients are fixed, and it is assumed that the DGs can supply the load demand. Since the output power of RESs is variable, Ref. [101] presents an adaptive droop and adaptive virtual impedance control strategy for microgrids with variable PV outputs and load demands to solve this issue. The droop coefficients are not fixed and are adjusted according to renewable energy variation, achieving proper sharing.

Some studies have focused on combining a virtual synchronous generator (VSG) and VI to improve the response to transient conditions. An adaptive virtual-impedance-based virtual synchronous generator (VSG) control approach is offered in Ref. [102] for grid-connected and islanded microgrids to decrease the impedance difference at the output of the inverter and enhance proportional reactive power sharing among DGs. The virtual impedance is designed based on the operation points of the microgrid and includes an adaptive virtual resistance and a fixed virtual inductance. This article uses a modified VSG control referred to as fuzzy-secondary-controller-based VSG control to implement adaptive virtual impedance control. Another hierarchical control scheme is introduced in

Ref. [103] and consists of primary and local control. The primary control is a higher level controller that acts in several seconds and is slower than the lower level one, which works in tens to hundreds of milliseconds. For this research, primary control consists of active and reactive power control (using power–frequency and power–voltage droop strategies) and virtual impedance (represented with virtual resistance and virtual inductance). The lower level controllers consist of virtual inertia control, voltage control, and current control loops. The output of the P–F droop is a reference power, which is used as an input for the virtual inertia controller, and the reference voltage is the output of the Q–F droop, which is the input of the virtual impedance. A nonlinear control algorithm based on control-induced time-scale separation and singular perturbation analysis is implemented to control the AC voltage and current and stabilize the system.

In most studies, the parameters of VI have been determined through small-signal stability; however, some researchers have offered optimization methods to find the optimized parameters for VI. The PSO algorithm is an optimization method that was utilized in Ref. [104] to optimize the virtual synchronous generator (VSG) and VI parameters in an islanded microgrid. For this purpose, a small-signal model for the microgrid was first developed. The parameters were initialized, and an optimization method and objective function were defined to find the optimized parameters. The results were compared with droop control and the combination of droop control and virtual impedance (droop control+VI) in the load change scenario. Moreover, the effect of virtual inertia and virtual damping on stability was investigated.

In addition to the previously mentioned applications, VI can work as a fault current limiter. An analytical small-signal model of a grid-forming voltage source converter is developed in Ref. [105]. The adaptive virtual impedance (VI) is deployed to limit the current of voltage source converters during grid faults. A three-phase fault is applied at the PCC point to tune the VI parameters to analyze them under the worst-case conditions. By considering the current and the X/R ratio limitations, VI parameters are formulated and tuned. A holistic controller parameter-tuning method is employed to guarantee small-signal stability. A new short-circuit current calculation algorithm is suggested in Ref. [106] for an islanded microgrid. To achieve optimal protection coordination, a virtual impedance fault current limiter is implemented. A two-stage optimal protection coordination is developed in this paper. The first stage is an optimal value for adaptive virtual fault current limiter calculation, and the second stage is finding the optimal setting for directional overcurrent relays. Another adaptive virtual impedance fault current limiter (VI-FCL) is offered in Ref. [107]. Unlike the previous method, different types of virtual impedance (including linear, exponential, and parabolic) and directional overcurrent relays (DOCRs) are deployed and compared. Then, optimal protection coordination for the islanded microgrid is investigated, and using the Interior-Point algorithm, optimal settings are obtained. The designed controller is tested in balanced and unbalanced conditions and considers the nonlinearity of the system.

Machine learning techniques are used in various areas of microgrid development. These techniques are advantageous compared to traditional methods as they do not require prior knowledge and can improve through environmental interaction. This means that system designers do not need to worry about the complexity and nonlinearity of the system. Due to its distinct features, RL is gaining popularity in different fields, such as microgrids (MG). In Ref. [108], the authors highlight the growing significance of the Internet of Things (IoT) and cloud computing in addressing industrial challenges. They specifically focused on efficient microgrid (MG) control, emphasizing the need for a robust and scalable Information Communication Technology (ICT) infrastructure. The study introduces a three-layer hierarchical control system for an inverter-based MG, incorporating cloud-based IoT infrastructure and a machine learning (ML) islanding detection scheme. In the primary layer, the paper implements voltage–frequency (V–F) droop control with virtual impedance for islanded mode and active–reactive (P–Q) power control for grid-connected mode. The secondary layer employs a decentralized averaging method to eliminate voltage

and frequency deviations, with communication facilitated through a lightweight IoT-based protocol using an edge device (ED). The ED incorporates a context-aware policy (CAP) to optimize communication traffic by comparing present and previous data values. A cloud-based ML model, specifically an artificial neural network (ANN), is developed in the tertiary layer for islanding detection. The ANN is trained using data from simulating islanding scenarios in Matlab, with phasor measurement unit (PMU) data communicated to the cloud for islanding prediction. A summary of VI strategies and their features is illustrated in Table 6.

Table 6. Summary of different types of virtual impedance.

Type of VI	Feature	Centralized/ Decentralized	Ref.
Complex VI	Control the power sharing	Decentralized	[88]
Inductive	Control the reactive power	Decentralized	[90]
Complex	Control the active and reactive power	Decentralized	[91]
Current-based, voltage-based	A hybrid active damping strategy	Decentralized	[92]
Frequency-based	Impact on harmonic and inverter stability	Decentralized	[93]
Current-based VI	Control the power sharing	Decentralized	[94]
Virtual flux droop	Control the power sharing	Decentralized	[95]
Virtual resistance/virtual inductance	Control of reactive power sharing	Centralized	[96]
Adaptive VI	Control of reactive power sharing	Centralized	[97]
Power-based VI	Control of reactive power sharing	Decentralized	[98]
Current-based VI	Modifying the virtual impedance for a negative sequence (NS) of the DG	Decentralized	[99]
Virtual resistance/virtual inductance	Control of reactive power sharing	Decentralized	[100]
Adaptive virtual impedance	Control of reactive power sharing	Decentralized	[101]
Adaptive virtual-impedance-based virtual synchronous generator (VSG)	Decrease the impedance difference at the inverter output; control of reactive power sharing	Decentralized	[102]
virtual impedance/virtual inertia control	Mitigate the impedance gap at the inverter output and enhance the proportional distribution of reactive power among DGs	Decentralized	[103]
Adaptive virtual impedance/virtual synchronous generator (VSG)	Alleviate the stability issues by imitating the synchronous generators	Decentralized	[104]
Adaptive virtual impedance-based current limitation	Limit the current of grid-forming voltage source converters (GFM-VSCs) during grid faults	Decentralized	[105]
An adaptive virtual impedance fault current limiter	Integrate as a supplementary control loop within the inverter control scheme to restrict fault currents	Decentralized	[106]
An adaptive virtual impedance fault current limiter	A two-stage optimization strategy is suggested to attain optimal protection coordination	Decentralized	[107]
Complex VI	Control the power sharing	Decentralized	[109]

5.2. Load Frequency Control

The instability of a power system in maintaining system frequency within the defined operational thresholds is called frequency instability. Generally, it is caused due to a significant imbalance between load and generation. Large frequency deviations cause damage to equipment, degrade load performance, overload transmission lines, and consequently, cause an unstable condition for the power system [110]. Load frequency control regulates power system frequency and is a major function of automatic generation control (AGC) [111]. The power system controls maintain the desired performance and stabilize the system in

the presence of disturbances. Power system control utilizes a control theory and technology, optimization methodologies, and expert or intelligent systems to enhance power systems' efficiency and capabilities under normal and abnormal operating conditions. Power system controls ensure stability and protect the system from hazardous incidents. Two main objectives of load frequency control are maintaining frequency and power interchanges with neighboring control areas at scheduled values. The area control error (ACE) is a control error signal, the real power imbalance between generation and load [112]. After filtering, the ACE is deployed as an input control signal for controllers such as the proportional–integral (PI) controller or PID. Figure 12 shows the schematic of the LFC in the two-area power system. $\delta PL1$ and $\delta PL2$ are the load changes. $R1$ and $R2$ are the droop resistances, and $B1$ and $B2$ are the droop coefficients. Tuning the controller settings is required to correct the probably accumulated frequency and net interchange errors due to integral control. Tuning the controller can play a crucial role in obtaining optimal LFC performance. Proper tuning of the controller parameters obtains good control and prevents excessive movement of units. Frequency control has become challenging and important due to the increasing size and complexity of interconnected areas and the growth in renewable energies. Due to the importance of the power system efficiency and reliability, maintaining the system frequency and tie-line flows closer to the scheduled value is required to fulfill the stability requirement. Therefore, in the modern power system, LFC has a crucial role in supporting power system exchanges and providing better conditions for electricity trading [111,113]. Figure 13 indicates different types of LFC that are investigated in detail.

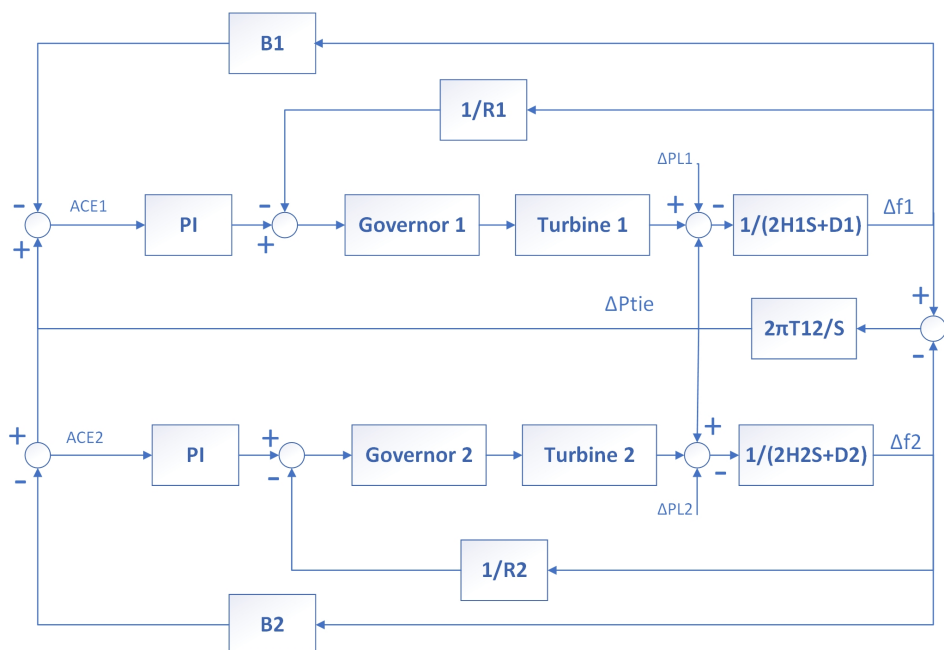


Figure 12. Schematic of LFC in two-area power system [114].

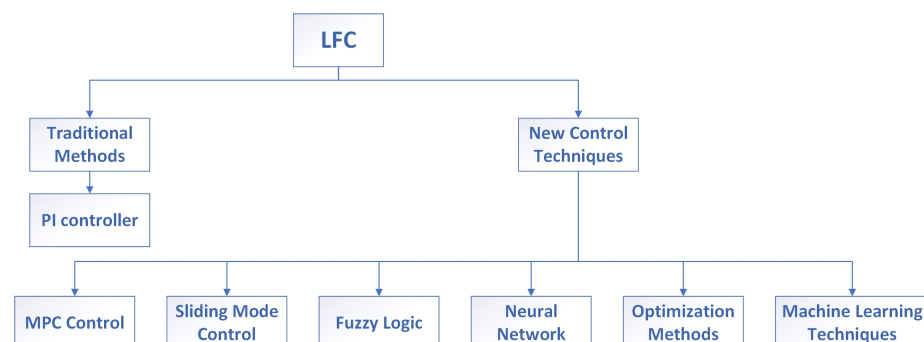


Figure 13. Various types of LFC.

5.2.1. Model Predictive Control (MPC)

In the past, the values of controller parameters were tuned based on experience and conventional methods such as Ziegler–Nicholas. These methods do not properly perform under random load variations and different operating conditions. Hence, many intelligent and optimized control methods have been proposed to improve the performance of load frequency control. Model predictive control is a researched method to improve load frequency control. The model predictive control (MPC) algorithm was proposed in Ref. [115]. The MPC algorithm has three main parts: prediction model, rolling optimization, and feedback correction. The prediction model predicts the future output of the controlled object using operational history data. Rolling optimization is performed online in contrast with traditional optimal control, which is offline. In rolling optimization, the MPC algorithm formulates an optimization problem over a limited future time horizon at each time step. This optimization problem aims to minimize a given cost function while identifying the best control inputs to give the system the desired trajectory. The prediction model of future projections, as well as the present condition of the system are considered during optimization. A rolling optimization is performed online, meaning the control inputs are continuously updated per the most recent measurements. The MPC algorithm can react quickly to alterations in the behavior of the system, disturbances, and changes in operating conditions thanks to its real-time adaptability. In addition, since rolling optimization solves smaller optimization problems at each time step compared with traditional offline optimization, it reduces the computational burden and has faster control responses. Finally, since there is a difference between predicted and actual values, the feedback correction mitigates this issue and compensates for future errors. In Ref. [116], the MPC controller is designed in the presence of wind farms and thermal power plants to improve LFC. Based on the analytical models of wind turbines and wind farms, the introduced frequency control strategy simplifies controller design and demonstrates improved frequency regulation ability compared to traditional control strategies. This method is applied to a multi-area hybrid system, enabling simultaneous control of wind farms and thermal power plants within the same area.

An adaptive model predictive control (AMPC) was proposed in Ref. [117] for a voltage source converter. This study uses superconducting magnetic energy storage (SMES), controlled by a supervisory adaptive model predictive control system, to mitigate the frequency control issues. Cascade control is designed for IGBT-based SMES to control the power exchange between the SMES and the power system. The reference power command is generated by a supervisory controller to fulfill the control objectives. A prediction of the energy level of the SMES is required, and AMPC restricts the SMES current within allowable limits. However, MPC-based controllers are too complex and are not extensively utilized in the industry.

5.2.2. Sliding Mode Control

Another control strategy for LFC is sliding mode control (SMC), which is one of the more powerful control methods and enhances control performance better than linear control strategies. Through this control method, all system states are initially directed towards the specified sliding mode surface before being guided to the origin along this surface. SMC is generally utilized with an embedded PI mechanism and can stabilize the power system within a broad range around the operating point. Event-triggered sliding mode control was proposed in Ref. [118] for multi-area power systems. This study initially proposes a new event-triggered discrete sliding mode surface function for each subsystem in a multi-area power system. Then, a sliding mode controller is designed to guarantee the stability of each subsystem and mitigate the effect of disturbances, including frequency fluctuation and load variation. This study is performed on three area power systems with a generator in each area. In Ref. [119], where just conventional energy is considered, an improved sliding mode control (SMC) is utilized as a basic controller; then, an adaptive dynamic programming strategy is employed to provide a supplementary

control signal to enhance the performance of LFC by considering the disturbances and uncertainties. The ADP-based supplementary controller is tuned online and can adaptively provide supplementary control signals by online weight updating for better adapting to uncertainties and real-time disturbances. But, in practical applications of this kind of control mode, there is an undesirable phenomenon of oscillations known as ‘chattering’. This phenomenon is harmful because it causes low control accuracy and high heat losses in power circuits. Chattering can occur due to two main reasons. The first reason is fast dynamics, which is neglected in the ideal model. The second one is the utilization of digital controllers with a limited sampling rate.

5.2.3. Fuzzy Logic Control

Fuzzy logic control has been utilized in different studies for LFC. Fuzzy logic control is an intelligence technique and employs prior experience of the functionary of the system. Decision-based rules are set up by the functionary based on analyzing the system behavior and input variables within the framework of the system. The inputs of fuzzy logic control are processed before generating outputs through three basic stages, including fuzzification, the decision-making stage, and defuzzification. The input variable is converted into a linguistic variable in the fuzzification stage, and the output of this stage is utilized to generate a fuzzified output. Fuzzy rules and input variables are used in the decision-making stage of fuzzy logic control to choose the best control action or output for the system. Fuzzy rules must be applied to the fuzzified input variables to determine how to modify the output of the system. In the defuzzification stage, at the end of the input processing stage, the fuzzified output is converted to the required output, which controls the system. The schematic of fuzzy logic control is shown in Figure 14 [120,121].

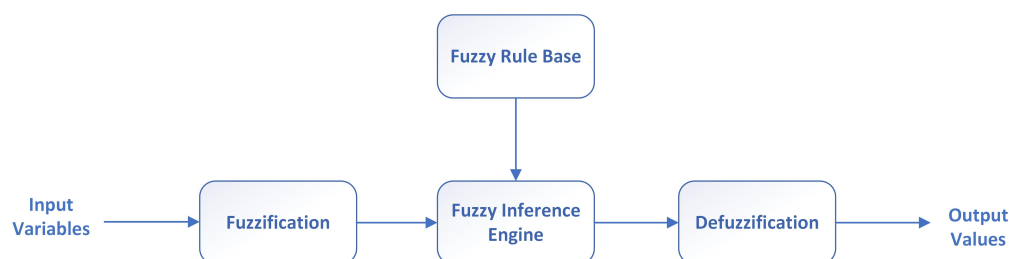


Figure 14. Fuzzy logic control process.

Ref. [122] proposes two layered fuzzy controllers, a pre-compensator and feedback fuzzy logic controller. The first is employed to generate and update the reference value for area control error (ACE), and the second reduces the steady-state error (ACE decay) to zero. This controller is proposed for two-area power systems with a thermal power plant in each area, and the results show that it has less overshoot and/or undershoot and less settling time than the conventional PI controller. However, they cannot be effective for multi-input problems. Ref. [123] proposes multi-area frequency control using a fuzzy generalized control method to tackle this issue. In this model, wind farms are integrated with power systems. The Takagi–Sugeno (T-S) fuzzy model of the interconnected power system incorporating wind farms (IPSWF) is initially developed. The primary objective is to achieve global linearization by employing local linear modeling. Then, the fuzzy generalized predictive control (Fuzzy-GPC) is modeled to forecast the output power of the IPSWF by making assumptions about future control actions. Because this approach involves online adjustment of the LFC parameters, it qualifies as an adaptive control technique. Fuzzy logic is robust and simple and can be modified according to our requirements. Despite its advantages, fuzzy logic control requires the exact model of the system to develop. Hence, this control strategy is complex and has high implementation costs. In addition, it is completely dependent on human knowledge and expertise. The efficiency of the controller is not high due to the inaccurate inputs and data.

5.2.4. Neural Network-Based Control Strategies

Neural network-based control methods have been utilized in industry to develop an advanced control algorithm for LFC. In Ref. [124], a PID neural network was designed to automatically identify and fine-tune the parameters of the controller system. The control algorithm in this research includes inputting the object value of the system into the controller and determining the weights of PID neural networks using the PSO algorithm. Then, the PID neural network is utilized to control the system, provide feedback on the output of the control system, and adjust the gains of the PID neural network by an improved gradient descent method. This process continues until the control error is small enough, and the process is terminated. An adaptive neural network Constrained control was proposed in Ref. [125] that is employed in a multi-area interconnected power system with hybrid energy storage to enhance LFC. The hybrid energy storage system contains a supercapacitor as the primary power source and a fuel cell as the secondary power source. Two types of neural networks are employed in this study. One of them is a Hammerstein-type neural network (HNN), and the other one is a PID-type neural network (PIDNN). The first one is designed to provide model information online by learning the power system with the hybrid energy storage system to update the PIDNN controller parameters. The PIDNN controller is a self-learning and self-adapting controller that provides the desired and appropriate power flow reference for the HESS to improve the LFC requirements. The challenge lies in training the ANN for a multi-area PSN due to load demand variations. To address this issue, the authors in Ref. [126] proposed particle swarm optimization (PSO) to optimize the ANN hidden layer nodes and initial neurons, resulting in a minimized mean-squared error. The study has an optimal LFC design using an artificial neural network (ANN) technique to restore frequency response during rapid load demand changes.

5.2.5. Optimization Methods

Some evolutionary computation methods utilize optimization methods to enhance load frequency control. Differential evolution (DE) was proposed in Ref. [127], which is an Evolutionary Algorithm (EA) originally designed for solving optimization problems over continuous domains. It is a kind of stochastic population-based evolutionary method. DE searches in stochastic ways to drive the population toward better solutions in the design space by applying mutation, crossover, and selection operators. It has only three control parameters. The performance of DE depends on the vector-generation scheme and choosing the control parameters. Furthermore, load frequency robust control for microgrids with energy storage systems using differential evolution was proposed in Ref. [128], in which an integrated power system with some renewable energies was studied. A robust method for designing the load frequency controller was employed. Then, the adaptive differential evolution algorithm was utilized to achieve the optimal weighting functions to obtain more robust performance.

Another soft-computing method for LFC is particle swarm optimization (PSO), a stochastic-based optimization algorithm. It depends on group effort and is based on the experience of all other community members, like fish or birds that move in a group. In other words, a bird flying and randomly searching for food can share its discovery and help the others in a community achieve the best hunt. A new PSO variant, STPSO, utilizes some statistical parameters as a tracker to find the global best value and decrease the number of irritations proposed in Ref. [127]. This method utilizes the statistical characteristics of society and facilitates exploring the global minimum for particles. This algorithm was employed to find optimized gains of PID controllers for four-area deregulated systems with different numbers of thermal power plants. In Ref. [129], a novel control method using PSO was proposed to improve the compensation precision and achieve better performance compared to traditional PI controllers. To prevent overcurrent and over-modulation faults in the shunt active power filter (SAPF), the paper proposes an enhanced current-limiting scheme utilizing particle swarm optimization (PSO). This scheme aims to simultaneously minimize Total Harmonic Distortion (THD) for the grid-side current and maximize the

utilization ratio for SAPF capacity. The proposed strategy offers flexibility and linearity in limiting control by calculating real-time optimal limiting ratios for each harmonic order.

The Firefly algorithm (FA) is a new metaheuristic and community-based technique developed in Ref. [130]. The flashing behavior of fireflies inspired the FA. In implementing this algorithm, based on the performance of an objective function, the flashing light is assigned, and randomly generated solutions are considered fireflies. If there is a brighter firefly, a firefly is attracted to that one and adjusts its position in that direction. The level of attraction is determined by the brightness difference between the two fireflies and some additional algorithmic parameters. If no brighter fireflies are nearby, the firefly moves randomly while searching the area for a better solution. In Ref. [131], a Firefly algorithm-based load frequency controller is developed in a two-area power system in the presence of thermal power and PV. The objective function is defined and consists of tie-line power and frequency deviation. The FA calculates PI controller parameters.

Some research works considered time delay in their studies and designed a controller. A PI-based controller is designed in Ref. [132] considering the transmission delay and sampling period. Initially, a delay-dependent LFC model of the power system is modeled based on sample data. Subsequently, a stability benchmark is formulated using the principles of Lyapunov theory. This controller is designed and tested for one area and three areas of a power system. In Ref. [133], LFC was investigated by considering multiple delays. In this regard, this study proposed a discrete-time model. Based on the region-division technique, a stability criterion was derived, and finally, a heuristic algorithm was employed for the proposed stability criterion to increase the computational efficiency.

5.2.6. Machine Learning-Based Control Strategies

Reinforcement learning (RL) is an artificial intelligence method and employs a learning method through interaction and feedback. It can be mainly categorized into model-free and model-based RL. Although RL is a promising approach to introducing flexible intelligence, the existing RL algorithms suffer from a few drawbacks that render them impractical to use in power systems. The model-free reinforcement learning (RL) methods need extensive exploration to discover optimal policies and long learning times, which is undesirable for a system with limited resources. Instead of random policy search, model-based RL can be used as an alternative [134]. Further, RL algorithms are often suggested in the discrete-time domain, perform effectively in low-dimensional dynamic systems, and can potentially lead power systems into unstable states. Therefore, an effective learning approach should be established on a model that captures the complexity, constraints, and nonlinearity of power systems. Ref. [135] introduced a multi-agent deep reinforcement learning technique, where each area within the multi-area power system employs its own frequency controller. The study presents an approach that involves both offline centralized learning and decentralized implementation. During the offline centralized learning phase, all agent parameters are updated, and controller parameters are optimized to enhance the performance of LFC. In addition, the generation rate constraints and generation dead band are considered physical constraints in this model. This RL method is tested in three interconnected areas and a 39-bus system in the presence of PVs and conventional energy resources. Since the controllers utilize local area state information, the results show that it can decrease the frequency deviation in the power system.

In Ref. [136], a load frequency control strategy based on Deep Q-learning (DQN) for islanded microgrids was developed to maintain stable operation under strong random interference and changes in network topology parameters. The authors first construct a load frequency control model for the microgrid, incorporating various energy. Next, they design the structure of the DQN-based frequency controller and define the state space, action space, and reward function. The optimal hyperparameters are obtained through adjustments. According to the simulation data, the DQN controller works better than the conventional PI and fuzzy control approaches, demonstrating the ability to learn from and replay past experiences. The DQN controller effectively deals with random microgrid load

frequency control issues and adapts well to complex operating conditions with variation in network topology parameters. Another novel approach to enhance load frequency control in multi-area interconnected power systems using a Deep Q-Network (DQN)-optimized Linear Active Disturbance Rejection Control (LADRC) controller was proposed in Ref. [137]. The DQN algorithm is combined with the ADRC controller to dynamically adjust the controller's parameters in real time, improving its adaptability to different disturbances in the power system. The effectiveness of the proposed method is demonstrated through simulations on three-area and four-area interconnected power systems. The DQN algorithm eliminates the need to divide the discrete state space, making it suitable for models with continuous or many states. However, the authors acknowledge that the action space still requires manual division, limiting the scope of the approach. Various types of LFC methods are summarized in Table 7, and the advantages and disadvantages of each method are investigated.

Table 7. Summary of different types of LFC, advantages and disadvantages.

Methods	Advantages	Disadvantages	Ref.
MPC	It is flexible and can handle a wide range of constraints and objectives; it is an effective way to deal with nonlinearities; this method is adaptable and can adapt to system changes.	The implementation and tuning of an MPC controller can be more complex; this method requires accurate power system models. Model inaccuracies or uncertainties can lead to poor control or even instability. It is based on solving optimization problems at each time step, has an extensive computational burden, and may not be suitable for real-time applications.	[115–117,138–140]
Sliding mode control	Fast response, robust to parameter variations, and can handle nonlinear systems.	Sensitive to modeling errors and uncertainties; it has a chattering issue, an undesirable phenomenon of oscillations, and complex implementation.	[118,119,141–145]
Fuzzy logic control	It can be adapted to learn from new data and can handle the nonlinearities of the system. It is flexible and can accommodate uncertainties and variations. This controller has a more stable response to sudden load changes or external interferences.	It has computational demands, which may affect real-time response, and it has yet to have universally accepted standards in its design and implementation, which are dependent on human knowledge and expertise.	[122,123,146–148]
Neural network-based control strategies	Suitable for nonlinear systems. They do not depend on the system parameters and can be trained based on past experience.	Complex in design and training; need sufficient training data; Overfitting and generalization issues when faced with new or unseen data, which can affect the performance of the controller.	[124–126,149]
Optimization methods	Efficient, adaptable, cost-effectiveness.	Depending on the system's parameters, they have Computational Complexity and are hard to implement in real time; it is hard to tune the controller.	[127,129,150–152]
Machine learning-based control strategies	These controllers are adaptable and can learn from historical data; they can handle nonlinear systems and have efficient control, and they can forecast system behavior and predict load changes. They can also optimize control policies.	Depend on the data quality; these controllers are complex due to Computational Complexity; they are hard to implement in real time and have generalization issues due to unforeseen scenarios in data for training the controller.	[134–137]

6. Future Work

In the modern landscape of power systems, the escalating integration of DERs has increased the prominence of control strategies. This article explores various levels of control strategies, with a focus on the essential need for advancements to enhance power system operation. The following outlines future work and research gaps in primary and secondary control strategies, particularly in the context of microgrids.

6.1. Future Work for Primary Control

6.1.1. Droop Control

Despite the evolution of control strategies, droop control remains a prevalent choice in power systems. Enhancing droop control performance is imperative, given the growth of DERs and their inherent lack of inertia. Several vital areas demand attention to achieve this:

- Robustness to variations in network parameters: Investigate and enhance the robustness of droop control mechanisms against variations in network parameters. Developing adaptive strategies will enable stability under diverse and dynamic conditions.
- Adaptation to advanced grid architectures: Explore adapting traditional droop control to accommodate advanced grid architectures, such as smart or microgrids. This adaptation ensures stability and optimal power sharing in evolving energy landscapes.

6.1.2. LVRT Strategies

For strategies related to low-voltage ride-through (LVRT), improvements are crucial to address challenges during extreme voltage dip situations:

- Real-time monitoring and diagnosis: Develop real-time monitoring and diagnosis tools during LVRT events. Utilize cutting-edge sensors and communication technologies to provide precise, timely information on voltage fluctuations, enabling proactive reactions.
- Hybrid control methods: Explore hybrid control methods that combine control system strategies with hardware modifications. This approach can effectively address extreme voltage dip situations, reduce system complexity, and mitigate the associated costs. In other words, For better LVRT performance, investigate combined strategies that include hardware adjustments and control system improvements. Examine the combination of adaptive control algorithms and upgraded hardware, like sophisticated protection devices, to enable successful LVRT in hybrid grids.
- LVRT enhancement for hybrid grids: Examine and create adaptable LVRT plans for hybrid grids. To ensure strong performance under a range of operating conditions, these strategies should be able to adapt to fluctuations in power generation and system configurations dynamically.

6.2. Future Work for Secondary Control

In the area of secondary control, virtual impedance and load frequency control (LFC) strategies necessitate advancements.

6.2.1. Virtual Impedance

- Sophisticated algorithms: Develop sophisticated virtual impedance algorithms that adapt to changing grid conditions. Investigate the application of artificial intelligence or machine learning to enhance the accuracy and flexibility of virtual impedance control.

6.2.2. Load Frequency Control (LFC)

- Diagnostic and real-time monitoring technologies: Employ diagnostic and real-time monitoring technologies to manage load frequency effectively. Explore ways to improve situational awareness and provide timely information for frequency regulation, leveraging advanced sensors and communication technology.

This comprehensive approach to future work underscores the need for adaptive, resilient, and technologically sophisticated control strategies in microgrid environments. Addressing these research gaps will enhance the stability and reliability of microgrid systems and contribute to the broader evolution of power systems in the era of DERs.

7. Conclusions

In conclusion, this review has explored the multifaceted landscape of inverter-based grid control strategies for DERs. Primary control strategies, including classic droop control and LVRT control strategies for PV and wind, were investigated. On the other hand, secondary control strategies, including virtual impedance (VI) and load frequency control (LFC), assume a crucial role in preserving grid stability and reliability and were discussed in detail. Looking ahead, the integration of DERs requires an understanding of these control strategies to address emerging challenges. As highlighted in the structured sections of this article, the comprehensive overview of DER topology, primary and secondary control strategies, and their operational features provides a foundation for future research and implementation. This review contributes to the broader discourse on sustainable energy by reviewing the advancements and challenges in inverter-based grid controls. Ultimately, the insights presented aim to propel the effective integration of DERs, facilitating a more robust and sustainable modern power grid.

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Abbreviations

The following abbreviations are used in this manuscript:

DERs	Distributed Energy Resources
DC	direct current
AC	alternating current
PCC	Point of Common Coupling
P/f	power–frequency droop
P/V	power–voltage droop
VBD	voltage-based droop
V/I	voltage–current droop
BESS	battery energy storage system
PFC	primary frequency control
SOC	state-of-charge
PV	photovoltaic
ESS	energy storage system
DFIG	Doubly Fed Induction Generator
LVRT	low-voltage ride-through
RSC	rotor-side converter
GSC	grid-side converter
WECS	Wind Energy Conversion System
SDR	series dynamic resistor
SDBR	Series Dynamic Braking Resistor

FCL	fault current limiter
SVC	Static Var Compensator
SATATCOM	Static Synchronous Compensator
UPFC	Unified Power Flow Controller
WTs	wind turbines
HVDC	high-voltage direct current
RPI	reactive power injection
MPPT	Maximum Power Point Tracking
PI	proportional–integral controller
VSG	virtual synchronous generator
VI	virtual impedance
LFC	load frequency control
AGC	automatic generation control
ACE	area control error
MPC	model predictive control
AMPC	adaptive model predictive control
SMES	superconducting magnetic energy storage
SMC	sliding mode control
PSO	particle swarm optimization
FA	Firefly algorithm
DE	differential evolution
EA	Evolutionary Algorithm
RL	reinforcement learning
DQN	Deep Q-learning

Appendix A

Table A1. Publications on primary and secondary control strategies from 2016 to 2023 studied in this review paper.

Droop Control	LVRT	Virtual Impedance	Load Frequency Control	Year of Publication
[39]	[80,81]	-	-	2016
[38]	[64]	-	[117,118,144]	2017
[29]	[77,82]	-	[125,128,131,142,150]	2018
[33,53,57,58]	[67]	-	[116,129,143,148]	2019
[23,50,51,56]	[73,74]	[96]	[134,135,145,152]	2020
-	[65,66,75]	[91,94,97]	[119,132,136–139]	2021
[26,36,43]	[83]	[88,90,93,95]	[126,133,141,146,151]	2022
-	[84]	[92]	[110,121,140,149]	2023

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